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Supply chain shocks and firm productivity: The role of reporting quality[☆]

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ABSTRACT

We examine the role of reporting quality in firms' responses to a supply chain shock. Extant research examines the role of financial or managerial reporting quality in recurring operating or investing decisions, but little evidence exists on its role in firm restructuring following adverse shocks. Our setting is the 1999 Taiwan earthquake, which disrupted supply chains for certain U.S. high-technology manufacturers. We find total factor productivity increases for affected firms relative to unaffected ones, and this effect is stronger when pre-shock reporting quality is higher. The relation between pre-shock reporting quality and post-shock productivity improvement is more pronounced for firms that incur post-shock restructuring charges or make larger spending cuts. Furthermore, post-shock accounting measures of performance improve more for affected firms with higher pre-shock reporting quality. Overall, our results suggest that accounting information plays a crucial role in enhancing firm resilience and adaptability in the face of external shocks.

1. Introduction

We study the relation between the quality of a firm's accounting and disclosure information and how productively it responds to a major shock to its operations. Considerable bodies of research have examined the role of financial reporting quality (FRQ) or managerial reporting quality (MRQ) in companies' operating and investing decisions. We are unaware, however, of research investigating whether reporting quality matters with respect to how well firms restructure after suffering a major, negative shock. Shocks can come in several forms: shifts in the competitive balance within an industry, supply chain disruptions that threaten an entire industry, and even shocks to the global economy such as the global financial crisis or the COVID-19 pandemic. How firms respond and restructure in the face of these shocks is of obvious importance, and having higher reporting quality (by which we mean both FRQ and MRQ) can help in selecting the most productive response.

A firm's reporting systems support managerial decision-making by reflecting the economic environment. However, reporting systems might struggle to provide relevant information when economic conditions shift suddenly. For example, in the face of a

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supply chain shock, inventory valuations might fluctuate with rising input costs, requiring adjustments to past inventory values for accurate future projections. This process can be more challenging if accounting systems are not of high quality and lack the flexibility to adapt. Consistent with this, prior research finds that poor inventory tracking and valuation can lead to lower inventory turnover and increased impairments (Feng et al., 2015). Also, accounts payable and receivable schedules assume normal payment cycles between suppliers and customers. A supply chain shock disrupts these timelines, leading to liquidity mismatches. Without accounting systems that can track changes in a timely manner, firms may misjudge their working capital requirements. In contrast, firms with high-quality, adaptable reporting systems, such as those incorporating real-time cost tracking or predictive analytics, are better positioned to make informed decisions even in the face of shocks.

To provide evidence on the role of reporting quality in firms' responses to shocks, we utilize a supply chain shock and study the linkage between pre-shock reporting quality and post-shock changes in firm productivity. Prior research shows there is wide dispersion in productivity across firms within the same industry (Hsieh and Klenow, 2009; Syverson, 2004, 2011). Information frictions have been proposed as a constraint impeding firms from converging quickly toward the productivity of their industry peers, and reporting quality is a natural candidate for such an information friction. Recent work has thus explored the role of accounting information in explaining the level of productivity at firms (Andreicovici et al., 2023) and in aggregate (Breuer, 2021; Choi, 2021), or the dispersion of productivity within industries (Barrios et al., 2019; Hann et al., 2020). We extend this literature by studying how reporting quality facilitates more productive responses under crisis conditions.

Productivity may increase in response to a supply chain shock if the shock leads to changes the firm was unwilling to make under less severe circumstances. Firms' managers and workers are often resistant to changing existing routines. The underlying reasons for such inertia can include: (i) doubts about the net benefits of a change, (ii) resource constraints that limit the investment needed for a change, (iii) limited managerial attention toward, or high uncertainty about, the effects of a potential change, and (iv) behavioral biases that favor the status quo (Gilbert, 2005; Hannan and Freeman, 1984; Longman et al., 2014). Crisis periods disrupt this inertia and provide an impetus to reduce slack and improve productivity to reduce the negative impact of the shock.¹

If higher reporting quality assists firms in making such productivity improvements, we should find that firms with better reporting quality in the period prior to the shock are able to achieve greater increases in productivity following it. To address our research question, we use the 1999 Taiwan earthquake, which adversely affected US high-technology manufacturing firms that sourced inputs (such as semiconductors and other electronic components) from Taiwan. The earthquake resulted in an increase in input prices for these firms, but not for competitors sourcing from alternative locations.

The increase in input prices resulted in higher production costs (or reduced output) for affected firms. High-technology manufacturing is a very competitive industry, and many US competitors were not harmed by the shock, so affected firms would have lost market share to rivals had they passed on the price increase to consumers (Ashenfelter et al., 1998; Gron and Swenson, 2000). In this sense, the supply chain shock increased the *threat* of competition for affected firms. Therefore, in response to the supply chain shock, affected firms have incentives to make changes to their operations. We thus begin our analyses by studying whether affected firms made any changes to their operating and investing activities. Our tests use a difference-in-differences design, where the control sample consists of industry peers that are unaffected by the shock. We find that following the earthquake, affected firms experienced an increase in restructuring charges, decreases in capital spending, R&D and SG&A, and no changes in acquisitions or their number of employees.

Operational changes in response to disruptions are expected to lead to enhanced productivity among affected firms. This is consistent with prior research that finds productivity increases for firms that adapt to competitive and various other shocks (Amiti and Konings, 2007; Hau et al., 2020; Lileeva and Trefler, 2010; Topalova and Khandelwal, 2011). Accordingly, we next study changes in the total factor productivity (TFP), a measure of how efficiently a firm converts inputs of labor and capital into outputs.

We find firm-level TFP increases 10%–11% for affected firms relative to the control sample following the shock. This translates to an increase of ten percentiles in productivity for affected firms, relative to control firms.² We include firm and year fixed effects to account for unobserved heterogeneity. Our results are robust to including multiple firm-specific control variables and to using an entropy-matched sample of control firms.

We next explore whether firms' preexisting reporting quality influences their ability to increase productivity following the shock. We use several proxies for the quality of firms' accounting systems and disclosure practices (hereafter referred to as "reporting quality"), including class action lawsuits related to financial reporting issues, weak disclosure controls, SEC investigations related to financial reporting or disclosure issues, fraud restatements, and the extent of supply chain disclosures in 10-K filings. Our reporting quality measures are intended to capture both general deficiencies in accounting quality and potential managerial intent to misstate information, both of which could weaken a firm's ability to generate decision-useful information.

We find that reporting quality significantly influences affected firms' TFP after the shock. When reporting quality is high, affected firms experience a significant increase in TFP relative to control firms, ranging from 11.3% to 12.6%. Firms with low reporting quality either see no increases in TFP relative to unaffected firms or witness a much lower increase in TFP (4.6% to 5%). The results are consistent across our measures of reporting quality. These findings show that better pre-shock accounting and disclosure practices correlate with greater improvements in post-shock productivity.

¹ See, for example, Amiti and Konings (2007), Hau et al. (2020), Lileeva and Trefler (2010), Topalova and Khandelwal (2011), Giroud and Mueller (2010, 2011) and Schmidt (1997).

² In comparison, Amiti and Konings (2007) find that a 10 percentage point decrease in input tariffs leads to a 12% productivity gain and Topalova and Khandelwal (2011) find that a 10 percentage point decrease in input tariffs leads to a 4.8% productivity gain.

High reporting quality can enhance firm productivity by ensuring effective information collection and dissemination within a firm (the MRQ portion of reporting quality). This allows managers to make better decisions under uncertainty, cutting slack and allocating resources more efficiently. Also, higher reporting quality enables external stakeholders, such as shareholders and creditors, to better monitor the firm's actions, particularly following operational disruptions, leading to improved managerial decisions (the FRQ portion of reporting quality). Both benefits of reporting quality suggest that when it is higher, the firm's responsiveness to crisis is enhanced, thereby boosting productivity. Our findings, which show increased restructuring and reduced tangible and intangible investments, suggest that high-quality information enables managers to cut overhead and allocate resources more efficiently.

Firm productivity and reporting quality are typically endogenous, as more productive firms tend to have higher reporting quality. In our setting, managers' incentives to increase productivity stem from firms' responses to an exogenous supply chain shock, with their ability to adapt depending on the preexisting quality of their accounting systems. As the shock occurs independently of firms' reporting quality, we can assess whether those with higher reporting quality respond more effectively. Additionally, because we measure reporting quality before the shock, other changes after the shock do not affect it.

It is important to consider that reporting quality may reflect one of many dimensions of managers' ability. High-ability managers tend to make multiple superior decisions that increase firm productivity (Bloom and Van Reenen, 2007; Bloom et al., 2013). High-ability managers are also likely to implement better reporting systems and management selection and reporting quality could be co-determined by other factors. Thus, firms' reporting quality can be viewed as one dimension of managerial ability. Recent research does, however, provide evidence that the quality of internal information systems has a separate effect from general management quality (Binz et al., 2023). We conduct additional analyses to tease apart the constructs of managerial ability and reporting quality by re-estimating our main analyses after controlling for a measure of managerial ability (based on Demerjian et al., 2012). Our results continue to hold, consistent with reporting quality capturing a significant construct distinct from managerial ability.

We provide further investigation of the channels by exploring whether the relation between reporting quality and TFP is concentrated among firms that face post-shock restructuring charges or make significant investment cuts. These firms, being most affected by the supply shock, are likely to leverage their information to maximize productivity. Our results confirm our hypothesis.

Our study provides several contributions. Addressing endogeneity is a key challenge in establishing a direct link between a firm's reporting quality and effective managerial decision-making. Prior research has approached this issue by examining spillover effects, where the reporting quality of one firm influences the economic decisions of peer firms, or by using variation in accounting standards, regulations, and enforcement efforts to assess their impact on decision-making (Roychowdhury et al., 2019). Instead, we focus on a setting where accounting standards and regulatory regimes remain constant, allowing us to isolate the impact of reporting quality on a firm's decision-making in response to an exogenous economic shock. Specifically, we analyze the supply chain crisis triggered by an earthquake, which creates a scenario where the quality of accounting information that firms generate and use in decision-making becomes crucial. By focusing on a key managerial objective of improving productivity amid heightened threat of competition, we show that firms with higher-quality reporting are better equipped to increase productivity under adverse conditions. These results highlight the critical function of accounting information in aiding managerial decision-making, especially under stress, and suggest that accounting information can be a crucial aspect of firm resilience and adaptability in the face of external shocks.

We also add to the literature linking reporting quality to investment efficiency. We find the productivity improvements in our setting largely arise from restructuring and reducing spending on investments in fixed assets, R&D and SG&A. We also find that the positive relation we document between reporting quality and TFP is concentrated among the companies that took post-shock restructuring charges or made significant cuts in their investments. Our findings thus indicate that higher reporting quality aids managers in adjusting investment toward more efficient levels when responding to a negative shock.

Our work also contributes to the nascent literature on the role of accounting information in explaining productivity by establishing a relation between firm-level productivity and measures of reporting quality. The only other research we are aware of linking accounting quality with firm-level productivity captures accounting quality by the intensity with which firms apply GAAP to map transactions into financial statements (Andreicovici et al., 2023). Most research aimed at linking accounting quality and productivity has examined either economy-wide (Choi, 2021) or industry-wide (Breuer, 2021; Hann et al., 2020), but not firm-level, productivity.

2. Related literature

Accounting research has long been interested in the role of reporting quality in management's decision-making, external monitoring, contracting, and internal and external capital allocation. Considerable work has therefore examined the role of reporting quality in enhancing operating efficiency (Chatterjee, 2023; Cheng et al., 2018; Cho, 2015; Feng et al., 2009, 2015). A large body of work also investigates the role of reporting quality in aiding investment efficiency (Roychowdhury et al., 2019), while some nascent research explores similar issues with regard to TFP.

However, this work focuses on the role of reporting quality when firms, industries and economies operate under normal circumstances. Limited prior work examines the role of reporting quality in firms' responses to shocks. Armstrong et al. (2019) find that aspects of reporting quality moderate both market pricing reactions and firms' future investment sensitivity to monetary policy surprises. In a similar vein, Balakrishnan et al. (2016) find that firms with more conservative accounting faced less underinvestment during the financial crisis. As opposed to a macro shock, we study how reporting quality influences firms' responses in the face of a supply chain shock.

We assess how well firms respond to a supply chain shock by using TFP as our primary outcome measure. We thus add to a growing body of work exploring the role of information frictions and reporting quality in explaining productivity. David et al.

(2016) present a theory of informational frictions that distort resource allocation across heterogeneous firms, leading to reduced aggregate productivity and output. Building on David et al. (2016) and Choi (2021) demonstrates that accrual accounting leads to more efficient resource allocation and increased aggregate productivity by improving managers' information about historical productivity.

Barrios et al. (2019), Breuer (2021) and Hann et al. (2020) relate FRQ to industry-wide resource allocation. Based on the FRQ measures of high quality accounting standards and external verification of financial reports, Barrios et al. (2019) find that financial reporting explains 10%–20% of the industry-wide dispersion in TFP. Arguing that external investors assess the productivity of the firm by studying the productivity of its assets, Hann et al. (2020) use asset informativeness and turnover quality as measures of financial reporting quality. They find these measures explain 8%–10% of the dispersion in TFP. Breuer (2021) examines the effect of country-level reporting and auditing mandates on the efficiency of industry-level resource allocation, and finds that while such mandates affect resource allocation, they do not appear to affect aggregate productivity. Other work utilizes changes in accounting standards to measure information quality, with Banker et al. (2021) exploiting the mandatory adoption of International Financial Reporting Standards (IFRS) to study its relation to country-level productivity. They find both TFP and labor productivity are positively associated with the adoption of IFRS.

As opposed to the industry or country level, we study changes in TFP at the firm level. To the best of our knowledge, the only other study to do so is Andreicovici et al. (2023), who propose that firms' metering problems are reflected in the intensity with which they apply GAAP to map economic transactions into financial statements. Metering is central to the theory of the firm as characterized by Alchian and Demsetz (1972), who describe the key problem of economic organization as finding cost effective ways of metering input productivity and rewards (p. 778). Andreicovici et al. (2023) use machine-learning to construct a firm-level, time-varying measure of accounting measurement intensity, which they then relate to firm growth, firm productivity, capital market activity, and contract terms. They find that higher accounting quality, as measured by accounting better capturing metering, is associated with higher TFP.

Our study builds on the limited extant research examining aspects of the connection between reporting quality and TFP. Whereas prior work has focused on relatively stable conditions, we study how reporting quality aids firms in responding to disruptive shocks.

3. Institutional background and setting

A 7.6 magnitude earthquake hit Chi-Chi, Taiwan, on September 21, 1999. The quake led to disruption of the supply of semiconductors and other electronic components manufactured in Taiwan, materials that are used in the production of high-technology devices, such as computers, mobile phones, medical devices, and other consumer electronics. Most of the semiconductor fabrication facilities and electronic-component manufacturers were located in the Hsinchu Science Based Industrial Park, 110 km north of the epicenter. This area included two of the largest manufacturers of semiconductors: Taiwan Semiconductor Manufacturing Company (TSMC) and United Microelectronics Corporation (UMC). Fig. 1 maps the location of manufacturing facilities, relative to the epicenter of the earthquake, and shows that many facilities were in areas that experienced tremors from the earthquake.³ These manufacturing facilities work with delicate precision machinery, so tremors from the earthquake disrupted the production process. The earthquake not only affected manufacturing facilities but also disrupted utilities for extended periods, leading to a supply shortage and an increase in the price of semiconductor wafers.⁴ The semiconductor industry was relatively unprepared for disruptions caused by an earthquake (Sherin and Bartoletti, 1999).

The earthquake adversely affected several US high-technology manufacturing firms that sourced materials, such as semiconductor wafers and other components, from Taiwan. In 1999, Taiwan was responsible for 17 percent of the world's supply of wafers and two-thirds of other computer components, such as motherboards, monitors, liquid-crystal displays, batteries, and chargers (Leachman and Leachman, 1999; Papadakis, 2006). The disruption to manufacturing facilities in Taiwan led to an increase in the price of memory and other electronic components, resulting in higher production costs for US manufacturers that relied on suppliers in Taiwan.

By the late 1990s, the semiconductor industry had evolved from a fully integrated model to one with an increasing reliance on foundries that handled chip manufacturing. At the beginning of 1999, 60% of foundries (by capacity) were located in Asia, 26% in the US, and 14% in Europe.⁵ Based on import data from the UN Comtrade database, presented in Appendix A, Taiwan accounted for 13.4% of US semiconductor imports in 1998, behind Malaysia (22.1%) and Japan (14.5%). In addition to chips manufactured abroad, the Comtrade data include imports of units that were assembled, tested, and packaged in other countries and explain why Malaysia accounts for the largest share of imports, even though the country had no foundries.⁶ Nevertheless, the data provide a lower bound for US reliance on semiconductor imports from Taiwan, and show a significant share of US semiconductor imports came from Taiwan.

³ Taiwan had 28 foundries at the time, and several facilities that manufactured electronic components. See "Picking up the pieces in Taiwan", *EE Times*, October 01, 1999.

⁴ The spot price of memory chips went up fivefold and contract prices went up by 25% (Papadakis, 2006). Fig. 2 shows the increase in memory prices following the earthquake. Such an increase is unusual as advances in manufacturing have resulted in a consistent decline in the prices of memory chips. A common explanation for the consistent price decline is Moore's law (Moore, 1965), which is the observation that the number of transistors in a chip doubles every two years, reducing the manufacturing cost per transistor and the cost of chips overall. Moore's law is used by the semiconductor industry to plan investments and set targets for research and development (Mulay, 2015).

⁵ As of 2015, only 13% of foundries were located in the US, and 76% were located in Asia including Japan, of which Taiwan's share was 24% (Source: IC Insights, Global Wafer Capacity).

⁶ Units that were manufactured in Taiwan's foundries but assembled, tested, or packaged in Malaysia and then imported to the US would show up as trade from Malaysia.

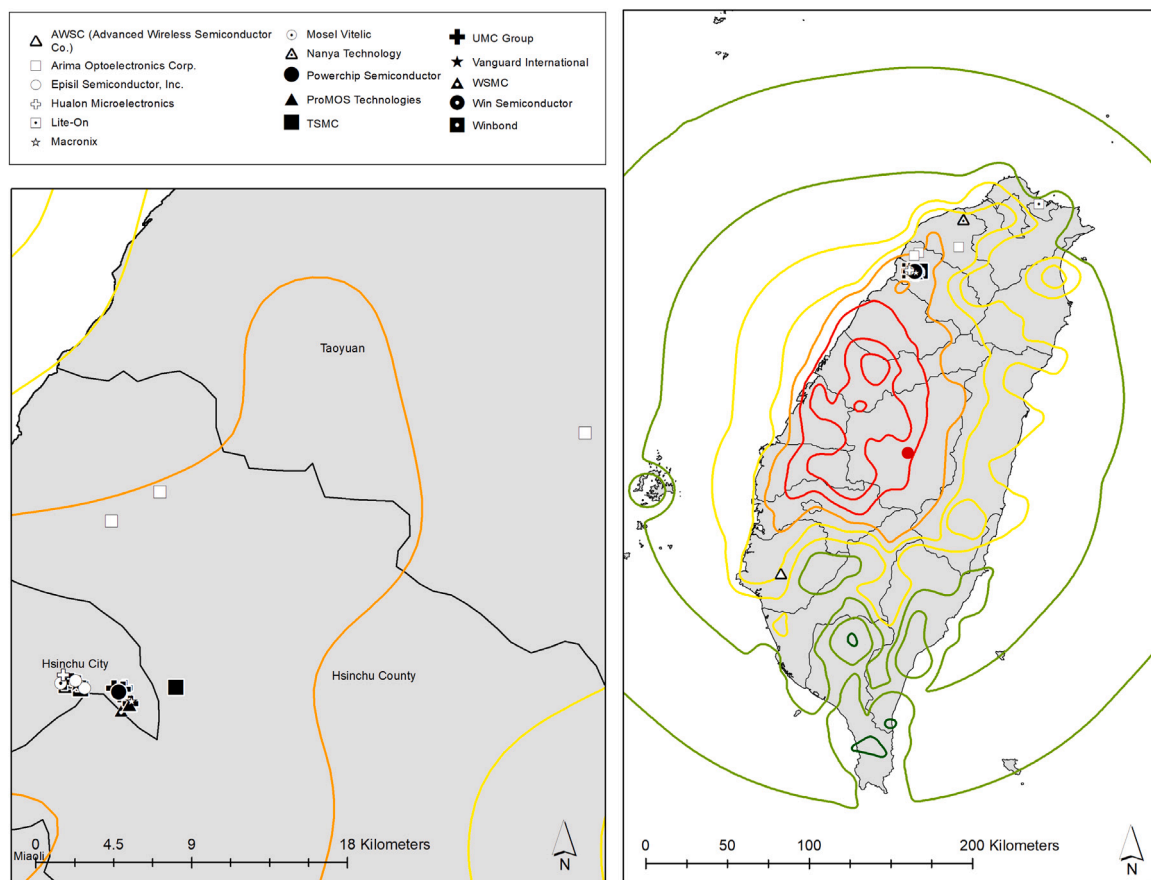


Fig. 1. Map showing earthquake intensity and location of manufacturing plants.

This figure shows locations of semiconductor foundries and component manufacturers, as of 1998–1999, relative to earthquake intensity, based on the Modified Mercalli Intensity Scale (<https://www.usgs.gov/programs/earthquake-hazards/modified-mercalli-intensity-scale>). Location data were compiled from company annual reports, Dataquest, SEMI, and company websites.

4. Data and sample

4.1. Sample of affected firms

We identify US high-technology manufacturing firms that were adversely affected by the supply-chain disruption caused by the earthquake by using regulatory filings and the Factset database. We search the text of SEC filings to identify firms that disclose ties to semiconductor chip suppliers in Taiwan. We complement the search with Factset, which consolidates firm-level information on supply-chain relations from regulatory filings and company websites. Our search focuses on the US firms' supplier relations before the earthquake. In some cases, firms reveal their connections to Taiwanese suppliers only while discussing the earthquake's impact on their operations. Therefore, we also use the filings from 1999 to infer supplier relations before the quake. The final sample consists of 103 unique US firms in high-technology manufacturing industries that have ties to suppliers located in Taiwan.

Appendix B provides details related to sample construction and a description of firms in the sample. Based on three-digit NAICS codes, almost all of the affected firms (98%) are classified under sub-sector 334: computer and electronic product manufacturing. Using three-digit SIC codes, affected firms are classified under codes 367 (electronic components and accessories — 68%), 357 (computer and office equipment — 18%), and 366 (communications equipment — 13%). These industries are a subset of industry number six (computers, software, and electronic equipment) in the Fama–French 12 industry classification.

The first disclosure related to the earthquake was made on September 30, or nine days after the earthquake, by Silicon Image Inc.⁷ The company stated the following in its filings:

⁷ Silicon Image provides video, audio, and data-connectivity solutions for the mobile, consumer electronics, personal computer, and enterprise markets (Source: Bloomberg).

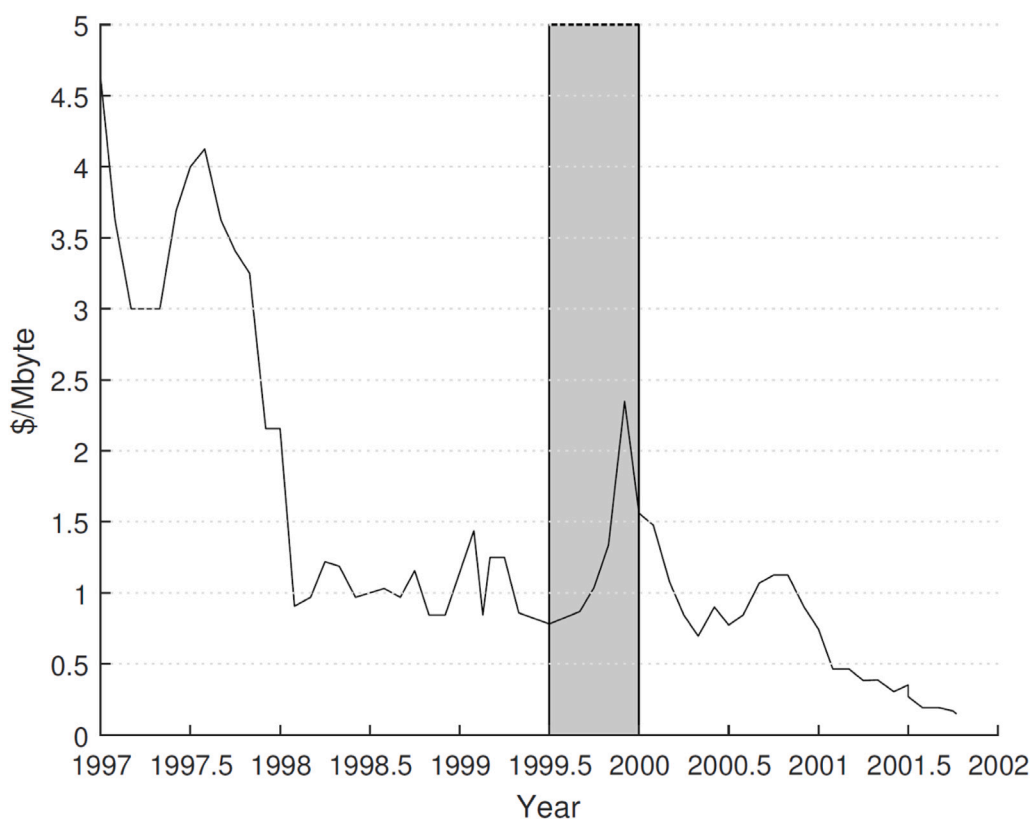


Fig. 2. Memory prices (1997 to 2001).

This Figure presents the evolution of the price (in US dollars) of a megabyte of memory from 1997 to 2001. The shaded region highlights the price rise after the 1999 earthquake in Taiwan. Prices are sourced from <http://www.jcmit.net/memoryprice.htm>.

We do not own or operate a semiconductor fabrication facility. We rely on Taiwan Semiconductor Manufacturing Company, an outside foundry, to produce all of our semiconductor products... We do not have a long-term supply agreement with Taiwan Semiconductor Manufacturing Company, or TSMC, and instead obtain manufacturing services on a purchase order basis. This foundry has no obligation to supply products to us for any specific period, in any specific quantity or at any specific price, except as set forth in a particular purchase order... we do not yet know the extent to which we will be adversely affected, if at all, by difficulties experienced by TSMC.

We also search firms' SEC filings for information on the types of contracts that affected firms had with suppliers in Taiwan in the year before the shock. Table 1 provides details related to these contracts. Of the firms that make disclosures related to their contracts, a majority rely on purchase orders. Purchase orders represent a single business transaction, are short-term in nature, and are usually sent to the supplier when the demand for the product arises. The short-term nature of the agreements between affected firms and their suppliers in Taiwan exposes these firms to short-term input price fluctuations due to the earthquake.

Table 2 presents descriptive statistics for firms adversely affected by the shock relative to other firms in the high-technology manufacturing sector. The control firms include SIC codes in industry number six in the Fama–French 12 industry classification (Computers, Software, and Electronic Equipment), excluding services (two-digit SIC = 73). As can be seen from the table, affected firms differ from unaffected firms (those not adversely affected by the earthquake) along several key dimensions. They are larger, more profitable, have higher market shares, have larger cash holdings and higher capital expenditures, but lower levels of inventory, leverage, and spending on selling, general and administrative expenses than firms that were not adversely affected by the earthquake. These differences are statistically significant. Table 2 also shows differences in pre-shock reporting quality measures between affected and unaffected firms. Whereas the two groups of firms are similar along most reporting quality measures, affected firms have a higher level of supply chain disclosures, potentially because they are larger. However, the difference is not significant when firm fixed effects are included (Fig. 7). We provide definitions for all variables in Appendix C.

4.2. Sample validation: One-day abnormal returns

We validate our sample construction by studying market returns for affected and unaffected firms around the day of the earthquake. Several media sources reported that stock prices of US producers that relied on chip foundries in Taiwan fell, whereas

Table 1
Contracts between affected firms and suppliers in Taiwan.

Contract type	Number of contracts	Percentage of firms	Percentage of firms excluding <i>Unspecified</i>
Purchase order	38	37%	68%
Contract manufacturing agreement	8	8%	14%
Joint venture	4	4%	7%
Take-or-pay	2	2%	4%
Supply agreement	2	2%	4%
Licensing agreement	1	1%	2%
Self operated	1	1%	2%
<i>Unspecified</i>	47	46%	
Total	103	100%	100%

This table shows the types of contracts that existed between US high-technology manufacturing firms that were adversely affected by the earthquake and their suppliers located in Taiwan. The data are sourced from SEC filings and the Factset database, and are for the year 1998.

Table 2
Descriptive statistics for affected firms versus unaffected firms.

	Affected Firms (1)		Unaffected Firms (2)		Difference in means (2)–(1)	
	Mean	Std	Mean	Std	Difference	t-statistic
<i>Firm characteristics:</i>						
Acquisitions	0.026	0.081	0.027	0.081	0.001	(0.085)
Capital expenditures	0.096	0.076	0.065	0.061	−0.031***	(−3.929)
Cash holdings	0.327	0.234	0.233	0.224	−0.094***	(−3.879)
Inventory	0.134	0.117	0.187	0.125	0.054***	(4.373)
Leverage	0.172	0.236	0.225	0.328	0.054*	(2.099)
Margin (gross)	0.292	0.963	0.274	0.909	−0.018	(−0.179)
Market Share	62.931	27.333	49.105	28.749	−13.826***	(−4.830)
Net debt	0.037	0.141	0.033	0.148	−0.004	(−0.273)
Net equity	0.163	0.374	0.132	0.357	−0.031	(−0.795)
R&D	0.200	0.227	0.165	0.200	−0.035	(−1.504)
ROA	−0.087	0.416	−0.193	0.504	−0.107*	(−2.407)
Sales	1.164	0.725	1.106	0.599	−0.058	(−0.783)
SG&A	0.203	0.266	0.328	0.290	0.125***	(4.476)
Size	5.040	2.119	3.866	2.052	−1.173***	(−5.331)
TFP	0.180	0.178	0.139	0.146	−0.041	(−1.828)
<i>Accounting quality measures:</i>						
Class Action Lawsuit	0.097	0.298	0.069	0.254	−0.028	(−0.924)
Weak Disclosure Controls	0.011	0.107	0.007	0.085	−0.004	(−0.343)
SEC Investigation	0.019	0.139	0.016	0.126	−0.003	(−0.240)
Restatement	0.019	0.139	0.016	0.126	−0.003	(−0.240)
Low Supply Chain Disclosures	0.203	0.406	0.527	0.500	0.323***	(5.605)
Observations	103		1000		1103	

This table presents financial characteristics for high-technology manufacturing firms that were adversely affected by the earthquake (Affected Firms) and those that were not (Unaffected Firms). Variables are measured in 1998, the year prior to the earthquake. Differences in characteristics are tested using *t*-tests of the difference in means. All variables are winsorized at the 1% and 99% tails of their respective distributions. Variable definitions are provided in [Appendix C](#). **p* < 0.1; ***p* < 0.05; ****p* < 0.01 (two-tailed).

the stock prices of competitors that sourced from other countries, such as South Korea, rose.⁸ The stock price reactions suggest market participants were aware of which firms would be adversely affected by the shock, based on the location of their suppliers.

As a check of the sample construction, we calculate one-day raw and abnormal returns for the affected and control samples. The earthquake occurred at 1:47 am local time on September 21, 1999, which was 17:47 UTC on September 20, 1999. Therefore when markets opened in the US on September 21, 1999, news of the quake was public knowledge, and market participants had the opportunity to react during the trading day. [Table 3](#) presents the one-day raw and abnormal returns for the affected and unaffected firms. The raw return for affected firms was −4.6%, compared to a raw return of −1.4% for the control sample. Abnormal returns (index-adjusted, CAPM-adjusted, and Fama–French three-factor model adjusted) for affected firms varied between −2.70% and −2.90%. Abnormal returns for control firms varied between −0.3% and 0.5%. The difference in means shows that returns for the affected firms were significantly more negative than those for the control firms — the difference in means ranges between 2.2%

⁸ See, for example, “Quake Rattles Chip Supply”, *CNN Money*, September 21, 1999; “Taiwan Quake to be Costly to World Technology Makers”, *The New York Times*, September 23, 1999.

Table 3
One-day market returns for high-technology manufacturing firms on day of earthquake.

	Affected Firms (1)		Unaffected Firms (2)		Difference in means (2)–(1)	
	Mean	Std	Mean	Std	Difference	<i>t</i> -statistic
Raw returns	–0.046	0.028	–0.014	0.046	0.032***	(8.606)
Index-adjusted returns	–0.027	0.028	0.005	0.046	0.032***	(8.606)
CAPM-adjusted returns	–0.028	0.031	–0.006	0.047	0.022***	(5.383)
Fama–French-adjusted returns	–0.029	0.030	–0.003	0.048	0.026***	(6.526)
Observations	79		559			

This table presents differences in one-day market returns for September 21, 1999 for firms affected and unaffected by the earthquake. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

Table 4
Changes in restructuring and investment.

	Restructure	Capital expenditures	R&D	Acquisitions	SG&A	Employees
	(1)	(2)	(3)	(4)	(5)	(6)
Affected $\times$ Post	0.094*** (3.208)	–0.023*** (–4.107)	–0.033*** (–2.605)	0.002 (0.516)	–0.067*** (–4.320)	–0.001 (–0.687)
Observations	11,862	10,791	10,975	10,975	10,975	9713
Adjusted R ²	0.388	0.442	0.597	0.119	0.560	0.225
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes

This table presents OLS estimates of changes in restructuring and investment for firms adversely affected by the earthquake (*Affected*) relative to those that were not, for 1994–2004. The indicator variable *Post* takes the value of 1 for 1999–2004. Standard errors are clustered at the firm level. To mitigate the effects of extreme observations, all continuous variables are winsorized at the 1% and 99% tails of their respective distributions in each sample year. All variables are defined in Appendix C. The *t*-statistics are presented in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

and 3.2%. The one-day abnormal returns provide a validation test for the sample construction and show that affected firms faced significantly negative abnormal returns relative to the control firms.

5. Empirical tests and results

5.1. Changes in restructuring and investment decisions

We hypothesize that disruptions in supply chains often compel firms to restructure or scale back their operations. Such changes could help firms adapt to the disruption by streamlining operations, reducing expenses, and reallocating resources, thereby reducing slack and improving productivity. Thus, we first examine post-shock changes in restructuring and investment by estimating the following difference-in-differences specification for 1994–2004,

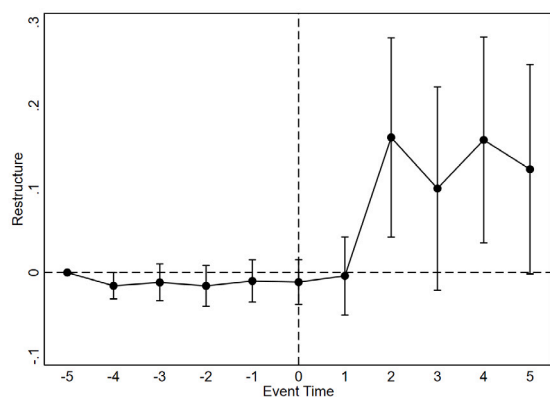
$$Outcome_{it} = \theta_0 + \theta_1 Affected_i + \theta_2 Post_t + \theta_3 Affected_i \times Post_t + \alpha_i + \delta_t + \epsilon_{it}, \quad (1)$$

where i indexes the firm, t the year, and *Outcome* is one of six outcomes described below. *Affected* is an indicator for firms that source from Taiwan and are therefore adversely impacted by the shock, whereas *Post* is an indicator for 1999–2004.⁹ The terms α and δ represent firm and year fixed effects, and ϵ_{it} is the error term. The main effects on *Affected* and *Post* are subsumed by the firm and year fixed effects.

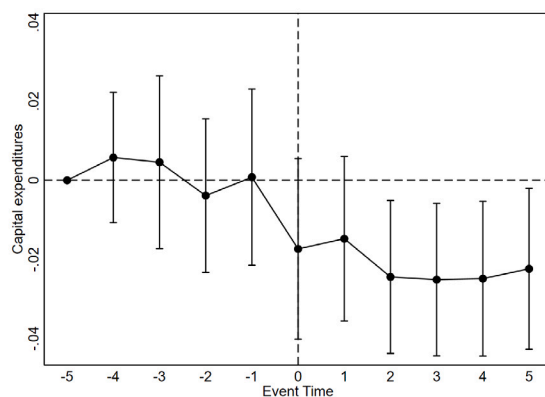
Outcome includes the following variables. *Restructure* is an indicator for whether firm i incurs restructuring costs in year t . The remaining five outcome variables are each for firm i in year t and are all scaled by the average of firm i 's year t and year $t - 1$ total assets. *Capital expenditures* is cash flow related to capital spending, *R&D* is research and development expense, *Acquisitions* is acquisition-related cash outflows (funds used for, and costs related to, acquisitions), *SG&A* is selling, general and administrative expense, and *Employees* is the number of employees.

Table 4 presents results from the estimations of Eq. (1). The column (1) results show that affected firms are 9.4% more likely to incur restructuring costs, relative to the control sample, following the shock. The estimates in columns (2), (3) and (5) indicate that affected firms make large post-shock cuts in capital expenditures, R&D and SG&A relative to the control firms. The per year cuts as

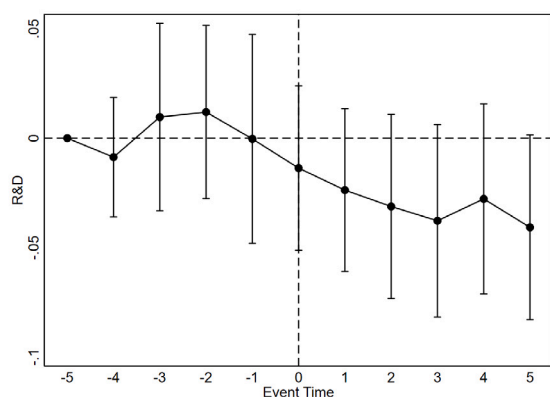
⁹ We include 1999 in *Post* because the shock occurred in September of that year. Therefore, some firms would have reacted to the shock in 1999 making it inappropriate to include this year in the pre-treatment period. Our inferences are unchanged if we redefine *Post* to include 2000–2004, or drop 1999 from the sample.



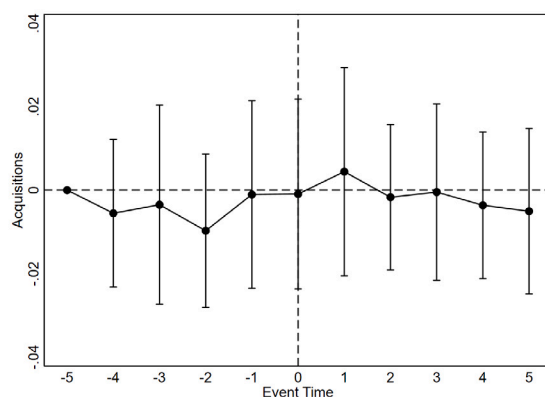
(a) Trend for restructuring charges in event time



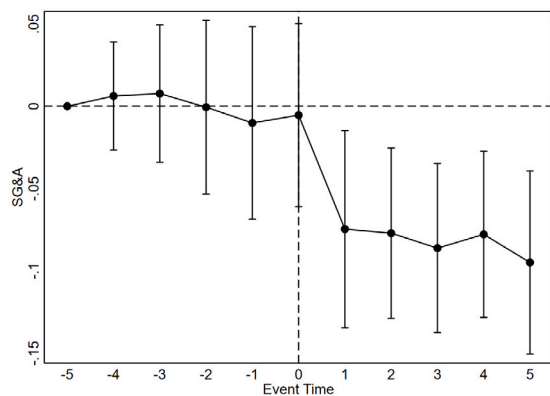
(b) Trend for capital expenditures in event time



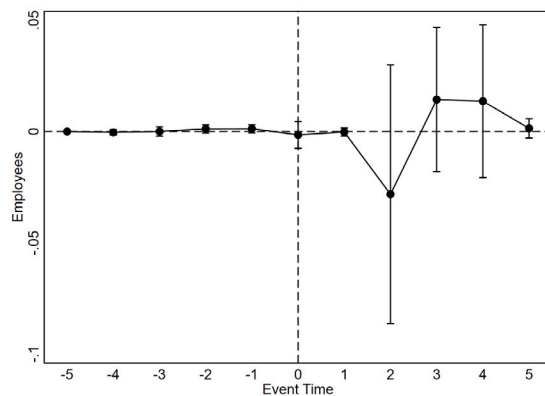
(c) Trend for R&D expenses in event time



(d) Trend for acquisitions in event time



(e) Trend for SG&A expenses in event time



(f) Trend for employees in event time

Fig. 3. Difference-in-differences trends for restructuring and investment variables in event time.

This figure presents trends in restructuring and investment in event time for the sample. The figure shows trends for the affected sample, relative to the control group of unaffected firms. The 95% confidence intervals are based on standard errors clustered by firm. Data are for the years 1994–2004.

a percentage of average assets for the affected firms relative to the control firms are 2.3% for capital expenditures, 3.3% for R&D, and 6.7% for SG&A. In contrast, the estimates in columns (4) and (6) show no significant differences between affected and control firms in their post-shock changes in acquisitions and employees.

Although affected and control firms vary based on some observable characteristics as shown in Table 2, we can generally draw valid inferences from a difference-in-differences analysis if the parallel trends assumption holds. To test for the presence of linear time trends, we estimate the following OLS model for 1994–2004:

$$Outcome_{it} = \gamma_t \times Affected_i + \gamma_i + \alpha_i + \epsilon_{it}, \quad (2)$$

where γ_t is an indicator for the year t . The remaining variables are as defined before.

Fig. 3 shows trends (i.e., coefficients of $\gamma_t \times Affected$) for each of the six outcome variables for the affected firms relative to the control sample. There are no significant differences in the outcome variable between affected and control firms in the years prior to the shock, supporting the reasonability of the parallel trends assumption that underlies our difference-in-differences specification. In addition, the plots show post-earthquake significant increases (decreases) in restructuring charges (capital expenditures and SG&A) for affected relative to control firms, consistent with our Table 4 inferences. There is also a considerable post-quake decrease in R&D for affected versus control firms, but in most post-quake years this difference is not quite significant at the 5% level. The results from Table 4 and Fig. 3 are consistent with increases in restructuring and reductions in investment by affected firms.

5.2. Changes in firm productivity

Our findings that affected firms cut investments and restructure are consistent with managers making operational adjustments in response to the shock, which will likely enhance productivity. High-tech manufacturing is competitive, with the top 50% of firms responsible for 55% of industry sales.¹⁰ In a competitive market, firms cannot pass on input price increases to consumers, especially if only a subset of firms are affected (Ashenfelter et al., 1998; Gron and Swenson, 2000). The shock would thus increase costs for affected firms due to higher prices for scarce inputs and the need to invest in alternative supply chain arrangements. Such shocks can align the interests of management and key stakeholders (Schmidt, 1997) and disrupt organizational inertia (Khrakovsky, 2024), making managers more likely to agree on productivity-enhancing but difficult decisions such as restructuring and terminating employees to cut organizational slack (Amiti and Konings, 2007; Hau et al., 2020; Lileeva and Treffer, 2010; Topalova and Khandelwal, 2011). If affected firms do not adapt, they will lose market share to competitors. In this sense, the shock increases the *threat* of competition.

To investigate whether affected firms witness an increase in firm-level productivity following the shock, we estimate the following difference-in-differences specification for 1994–2004:

$$TFP_{it} = \theta_0 + \theta_1 Affected_i + \theta_2 Post_t + \theta_3 Affected_i \times Post_t + \gamma X_{it-1} + \alpha_i + \delta_t + \epsilon_{it}, \quad (3)$$

where i indexes the firm, t the year, and TFP is total factor productivity. X is a vector of lagged control variables and includes *Acquisitions*, *Capital expenditures*, *Cash holdings*, *Inventory*, *Leverage*, *Margin(gross)*, *Net debt*, *Net equity*, *R&D*, *ROA*, *Sales*, *SG&A* and *Size*. The remaining variables are as defined previously. The main effects on *Affected* and *Post* are subsumed by the firm and year fixed effects. The estimate of θ_3 will be positive if TFP for affected firms increases relative to control firms following the supply chain shock.

Total factor productivity is a measure of how efficiently a firm converts labor and capital inputs into outputs. To calculate it we follow the method developed in Levinsohn and Petrin (2003), which takes into account the correlation between unobservable productivity shocks and input levels. This method has also been used in recent accounting work to measure TFP (e.g., Hann et al., 2020). We discuss details related to the estimation in Appendix D.

Table 5 presents the results from the estimation of Eq. (3), where TFP is the natural logarithm of one plus total factor productivity estimated based on Levinsohn and Petrin (2003). Columns (1) and (2) show that post-shock productivity increased significantly for affected firms relative to the control firms. Column (1) does not include firm-level control variables whereas column (2) does. To assess the robustness of these results, in column (3) we match affected and control firms on observable characteristics presented in Table 2. We use entropy balancing to achieve covariate balance based on the first two moments using data from the year prior to the shock (Hainmueller, 2012). Our results continue to show a significant increase in the productivity of affected firms relative to control firms. In terms of economic magnitude, TFP increases 10%–11% for affected firms relative to the control sample following the shock.¹¹ Finally, for ease of interpretation, in column (4) the dependent variable is percentiles of TFP. Relative to the matched controls, affected firms' productivity is ten percentiles higher in the years following the shock compared to the period prior to it.

To test for the credibility of the parallel trends assumption, we estimate the following OLS model for 1994–2004:

$$TFP_{it} = \gamma_t \times Affected_i + \gamma_i + \alpha_i + \epsilon_{it}, \quad (4)$$

where the variables are as defined before. Fig. 4 presents a plot for the coefficients of $\gamma_t \times Affected$, where the benchmark year is 1994. The figure shows no time trend prior to the shock, and a distinct increase in TFP for affected firms following the shock. The year of the shock is 1999, which is set as *Event Time* = 0. However, because the shock occurred during 1999, there is a slight increase in TFP in that year for affected firms relative to control firms. Fig. 5 presents the conditional trendlines separately for

¹⁰ The high-tech manufacturing industry has a low Herfindahl index of 136.6, relative to a mean of 166 for all manufacturing industries. These figures are sourced from the 1997 US Census Bureau's Concentration Ratios in Manufacturing. The Census Bureau's concentration measures include data from private firms and thus provide a complete picture of industry competition.

¹¹ These magnitudes are calculated as follows: $(e^{0.104} - 1) \times 100 = 11$. This is illustrative of the approach we use in all subsequent analogous cases.

Table 5
Changes in total factor productivity.

	TFP	TFP	TFP (entropy matched)	TFP ranks (entropy matched)
	(1)	(2)	(3)	(4)
Affected $\times$ Post	0.104*** (4.379)	0.103*** (4.455)	0.096*** (4.027)	10.521*** (3.782)
Acquisitions		0.018 (0.704)		
Capital expenditures		−0.250*** (−4.799)		
Cash holdings		0.068** (2.328)		
Margin (gross)		0.114*** (2.805)		
Inventory		−0.003 (−0.046)		
Leverage		−0.012 (−0.975)		
Net debt		0.037** (2.318)		
Net equity		0.029*** (3.522)		
R&D		−0.312*** (−4.309)		
ROA		0.010 (0.680)		
Sales		0.062*** (5.474)		
SG&A		−0.028 (−0.904)		
Size		−0.018*** (−3.013)		
Observations	7250	7152	6578	6578
Adjusted R ²	0.832	0.844	0.844	0.891
Controls	No	Yes	No	No
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

This table presents OLS estimates of changes in TFP for firms adversely affected by the earthquake (*Affected*) relative to those that were not, for 1994–2004. The indicator variable *Post* takes the value of 1 for 1999–2004. The dependent variable in columns (1)–(3) is TFP calculated based on Levinsohn and Petrin (2003). The dependent variable in column (4) is percentile ranks of TFP. Columns (1) and (2) include the full sample of firms, whereas columns (3) and (4) include entropy matched samples (matched on observable characteristics on the first two moments). The control variables in column (2) are lagged by one year. Standard errors are clustered at the firm level. To mitigate the effects of extreme observations, all continuous variables are winsorized at the 1% and 99% tails of their respective distributions in each sample year. All variables are defined in Appendix C. The *t*-statistics are presented in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

affected and control firms and shows that the affected firms drive the post-shock increase in TFP. Overall, the results in Table 5, Figs. 4, and 5 indicate that affected firms witness increases in their productivity, suggesting that they respond to the supply chain shock by reducing slack.

We next tie the productivity gains to changes in post-shock operating activities. Motivated by Table 4 results, we assess whether firms that cut investments or incurred restructuring charges increased their productivity more by estimating the following OLS model:

$$TFP_{it} = \theta_0 + \theta_1 Affected_i + \theta_2 Post_t + \theta_3 Take\ Action_i + \theta_4 Affected_i \times Post_t + \theta_5 Take\ Action_i \times Post_t + \theta_6 Affected_i \times Take\ Action_i \times Post_t + \gamma X_{it-1} + \alpha_i + \delta_t + \epsilon_{it}, \quad (5)$$

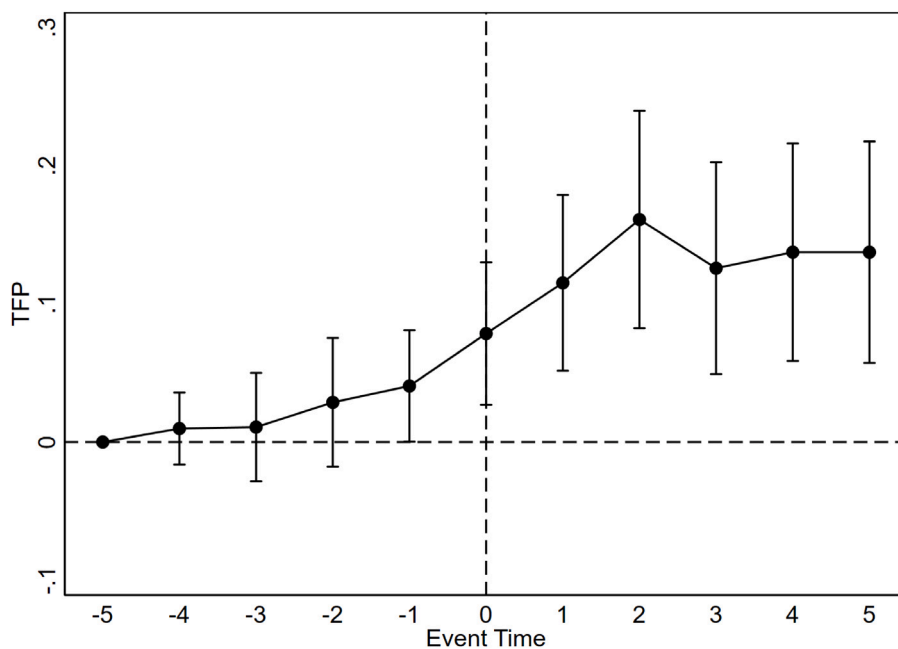


Fig. 4. Difference-in-differences trends for TFP in event time.

This figure displays the evolution of total factor productivity (TFP) in event time for the sample. The figure shows trends for the affected sample, relative to the control group of unaffected firms. The 95% confidence intervals are based on standard errors clustered by firm. Data are for the years 1994–2004.

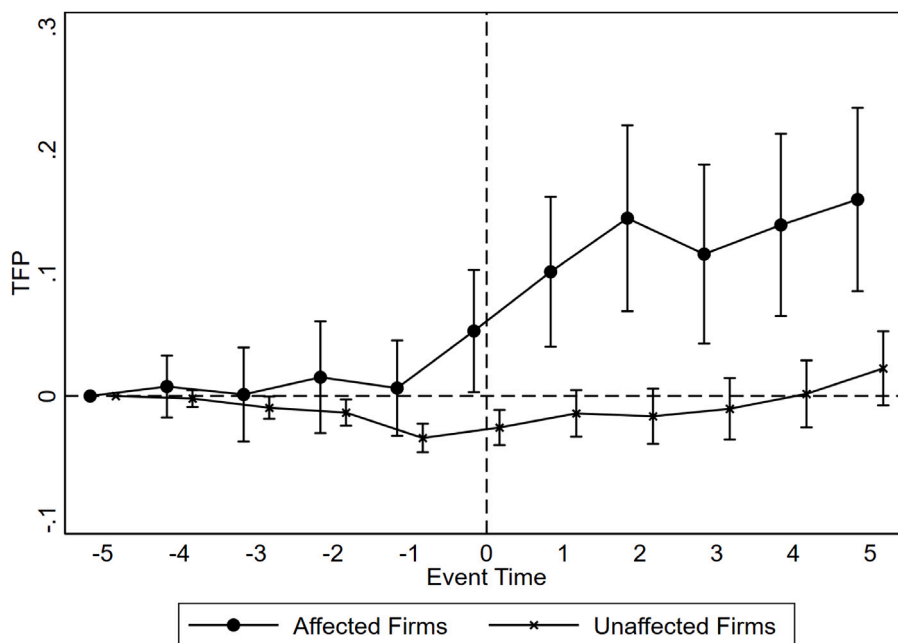


Fig. 5. Trends for TFP in event time.

This figure displays the evolution of total factor productivity (TFP) in event time for the sample, separately for affected and unaffected (control) firms. The 95% confidence intervals are based on standard errors clustered by firm. Data are for the years 1994–2004.

where *Take Action* is an indicator for firms that restructure or cuts investments post-shock. The remaining variables are as defined before. The main effects on *Affected*, *Take Action*, and *Post* are subsumed by firm and year fixed effects. We expect θ_6 to be significantly positive if restructuring or reducing investment is associated with increases in firm productivity.

Table 6
Post-shock actions and total factor productivity.

	TFP	TFP
	(1)	(2)
Affected $\times$ Post	0.036 (1.231)	0.037 (1.255)
Take Action $\times$ Post	−0.001 (−0.077)	0.007 (0.903)
Affected $\times$ Take Action $\times$ Post	0.085** (2.511)	0.079** (2.345)
Acquisitions		−0.023* (−1.744)
Capital expenditures		−0.228*** (−6.933)
Cash holdings		0.036*** (2.692)
Margin (gross)		0.108*** (4.909)
Inventory		−0.031 (−1.241)
Leverage		−0.006 (−1.100)
Net debt		0.015** (1.973)
Net equity		0.016*** (3.455)
R&D		−0.154*** (−5.535)
ROA		−0.001 (−0.100)
Sales		0.041*** (7.100)
SG&A		−0.023 (−1.615)
Size		−0.010** (−2.396)
Observations	6899	6802
Adjusted R ²	0.911	0.919
Controls	No	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes

This table presents OLS estimates of changes in TFP for firms adversely affected by the earthquake (*Affected*) relative to those that are not. The indicator variable *Take Action* equals one if the firm restructures or reduces investment in the period following the shock, and zero otherwise. The indicator variable *Post* takes the value of 1 for 1999–2004. The dependent variable (TFP) is calculated based on [Levinsohn and Petrin \(2003\)](#). The control variables in column (2) are lagged by one year. Standard errors are clustered at the firm level. To mitigate the effects of extreme observations, all continuous variables are winsorized at the 1% and 99% tails of their respective distributions in each sample year. All variables are defined in [Appendix C](#). The *t*-statistics are presented in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

We present results from the estimation in [Table 6](#). Affected firms that restructure or reduce investments following the shock witness an 8.2%–8.9% increase in productivity relative to affected firms that do not take such actions. For affected firms that do not restructure or cut investments, TFP is not significantly different from that of unaffected firms in the post-shock period (given the insignificant coefficients for *Affected* $\times$ *Post*). These results suggest that productivity gains materialize due to operational changes. Such operational changes are more likely to occur after shocks that force management to take actions they otherwise would not.

Table 7

Reporting quality and total factor productivity.

Panel A: Correlation of reporting quality measures					
	Class Action Lawsuit	Weak Disclosure Controls	SEC Inves- tigation	Restate- ment	Low Supply Chain Disclosures
Class Action Lawsuit	1.000	−0.030***	0.220***	0.279***	−0.095***
Weak Disclosure Controls	−0.034***	1.000	−0.013	−0.014	−0.046***
SEC Investigation	0.200***	−0.012	1.000	0.472***	−0.043***
Restatement	0.344***	−0.014	0.579***	1.000	−0.069***
Low Supply Chain Disclosures	−0.089***	−0.046***	−0.022	−0.093***	1.000

Panel B: Reporting quality and total factor productivity					
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Weak Disclosure Controls	Low Reporting Quality = SEC Investigation	Low Reporting Quality = Restatement	Low Reporting Quality = Low Supply Chain Disclosures
	TFP	TFP	TFP	TFP	TFP
	(1)	(2)	(3)	(4)	(5)
Affected × Post	0.119*** (7.006)	0.114*** (6.795)	0.107*** (6.691)	0.107*** (6.690)	0.115*** (5.716)
Low Reporting Quality × Post	0.003 (0.197)	−0.073** (−1.978)	−0.043** (−1.981)	−0.043* (−1.667)	−0.001 (−0.077)
Affected × Low Reporting Quality × Post	−0.081** (−2.014)	−0.077* (−1.866)	−0.062** (−2.346)	−0.062** (−2.048)	−0.066** (−2.356)
Sum of coefficients (<i>F</i> -statistic)	0.038 (1.13)	0.037 (1.02)	0.045** (4.24)	0.045* (3.02)	0.049** (6.08)
Observations	6802	5441	6803	6803	4061
Adjusted R ²	0.919	0.920	0.919	0.919	0.879
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

Panel C: Reporting quality and total factor productivity (controlling for managerial ability)					
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Weak Disclosure Controls	Low Reporting Quality = SEC Investigation	Low Reporting Quality = Restatement	Low Reporting Quality = Low Supply Chain Disclosures
	TFP	TFP	TFP	TFP	TFP
	(1)	(2)	(3)	(4)	(5)
Affected × Post	0.118*** (6.935)	0.114*** (6.790)	0.106*** (6.641)	0.106*** (6.643)	0.115*** (5.697)
Low Reporting Quality × Post	0.007 (0.447)	−0.077** (−2.198)	−0.041** (−2.147)	−0.041* (−1.655)	−0.001 (−0.114)
Affected × Low Reporting Quality × Post	−0.083** (−2.111)	−0.076* (−1.922)	−0.064*** (−2.616)	−0.064** (−2.203)	−0.066** (−2.367)
Managerial Ability	0.088*** (4.794)	0.093*** (4.497)	0.090*** (4.858)	0.088*** (4.775)	0.072*** (3.397)
Observations	6705	5398	6708	6708	3991
Adjusted R ²	0.920	0.922	0.919	0.919	0.880
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

This table presents correlation (between the various measures of reporting quality (Panel A) and OLS estimates of changes in TFP for firms adversely affected by the earthquake (*Affected*) relative to those that are not (Panel B). Panel C reproduces Panel B results after controlling for the managerial ability measure of Demerjian et al. (2012). In Panel A, Pearson coefficients appear above the diagonal and Spearman coefficients appear below the diagonal. In Panel B and Panel C, the indicator variable *Post* takes the value of 1 for 1999–2004. The dependent variable (TFP) is calculated based on Levinsohn and Petrin (2003). Controls are lagged by one year and include: *Acquisitions*, *Capital expenditures*, *Cash holdings*, *Inventory*, *Leverage*, *Margin(gross)*, *Net debt*, *Net equity*, *R&D*, *ROA*, *Sales*, *SG&A* and *Size*. The *F*-statistic tests if the sum of coefficients on *Affected* × *Post* and *Affected* × *Low Reporting Quality* × *Post* equals zero. Standard errors are clustered at the firm level. To mitigate the effects of extreme observations, all continuous variables are winsorized at the 1% and 99% tails of their respective distributions in each sample year. All variables are defined in Appendix C. The *t*-statistics are presented in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

5.3. Reporting quality and firm productivity

Our results thus far indicate that affected firms make operational changes and experience increased productivity relative to control firms in the period following the shock. We next examine whether reporting quality influences firms' ability to increase their productivity. Firm productivity and reporting quality are generally endogenous because more productive firms are also likely to have higher reporting quality. Our setting provides identification advantages that allow us to isolate the impact of reporting quality on firm productivity. We provide a framework in [Appendix E](#) that clearly outlines the empirical advantages of our setting.

Specifically, the supply chain shock does not directly affect productivity or reporting quality. Instead, the increase in firm productivity is the result of a response by firms to the exogenous supply chain shock, where firms' ability to respond depends on the predetermined quality of their accounting systems (captured by the parameter θ in the framework). We can determine whether firms with higher preexisting reporting quality respond more effectively to the shock because the shock itself is unaffected by firms' accounting practices. Also, we measure accounting quality before the shock; therefore, other changes after the shock do not influence it.

We capture reporting quality by utilizing five different measures of accounting and disclosure quality. We use multiple measures because our aim is to examine various aspects of firms' accounting information and disclosure practices. Our first measure, *Class Action Lawsuit*, is an indicator that takes a value of one if the beginning of the class period in a securities class action or collective action lawsuit related to financial reporting occurred in the five years before the shock. Our second measure *Weak Disclosure Controls*, is an indicator that takes a value of one if the firm was revealed to have weak disclosure controls under Section 302 of the Sarbanes–Oxley (SOX) Act of 2002. We utilize disclosures under Section 302 instead of Section 404 because SOX 302 disclosures were effective much earlier (August 29, 2002). SOX 404 reporting requirements were effective November 15, 2004, but only for accelerated filers. Furthermore, some firms might have reduced their potential internal control issues that would have been reported under Section 404 shortly before Section 404 became effective for them. Therefore, Section 302 disclosures are more likely to reflect conditions that existed in 1998 and yield a larger sample.

Our third measure, *SEC Investigation*, is an indicator that takes a value of one if the SEC started an investigation related to financial reporting or disclosure issues against the firm in the five years prior to the shock. The investigation categories we utilize relate to books/records/controls; delinquent filings; financial fraud/issuer disclosures; internal audit/internal controls; and unusable records. We obtain these SEC investigation data from [Holzman et al. \(2024\)](#), who acquired the SEC investigation dates and classifications via FOIA requests, and manually identified corresponding Compustat identifiers (GVKEYs). [Holzman et al. \(2024\)](#) also credit [Blackburne et al. \(2021\)](#) in the matching process, as their data were cross-referenced to ensure consistency and maximum sample size.

Our fourth measure, *Restatement*, is an indicator which equals one if the firm had an intentional misstatement (i.e., an irregularity) in the years prior to the shock, identified using the data and methodology of [Hennes et al. \(2008\)](#). These restatement data begin in 1997. Specifically, [Hennes et al. \(2008\)](#) classify restatements as intentional if any one of the following three criteria are met. First, restatements that mention “fraud” or “irregularity” are labeled as intentional misstatements. Second, those associated with SEC or DOJ investigations are classified similarly. Last, restatements linked to independent investigations are also deemed irregularities.

Our fifth and final measure is related to the quality of supply chain disclosures in the period prior to the shock. We utilize the disclosures in items 1 and 7 of firms' 10-K filings and a semi-supervised machine learning approach to measure the extent to which firms disclose supply chain-related information.¹² The degree to which managers openly discuss their supply chain issues in 10-K filings is likely an indicator of their awareness and proactive management of these issues. This transparency with investors regarding supply chain challenges may also reflect their preparedness to respond promptly and effectively to disruptions. The commitment to high-quality external disclosures can foster managerial learning from the internal information gathering needed to support the disclosures. Managers can then incorporate this knowledge into operational and investment decisions ([Roychowdhury et al., 2019](#)).

To create the supply chain disclosures measure, we use a neural network to detect relevant words and their synonyms based on the context provided by neighboring words, encoding each unique word as a vector of numbers. This process is facilitated by a word-embedding model, which helps in recognizing words with similar meanings based on the principle that synonymous words tend to appear in similar contexts ([Mikolov et al., 2013](#)). Prior work has utilized this methodology to quantify aspects of firms' corporate culture ([Gallo et al., 2023](#); [Li et al., 2021](#)), and the content of bank enforcement actions ([An et al., 2024](#)).¹³

To apply the machine-learning method, we introduce the following initial seed words: “supply”, “supplier”, “supply chain”, “procurement”, and “procure”. We then train a *Word2Vec* model to identify word vectors associated with these seed words and then manually review the words produced by the algorithm to verify that our dictionary contains relevant words.¹⁴ We then score each 10-K by using the weighted count of the number of words associated with supply chain disclosures (based on our dictionary) divided by the total number of words in the 10-K items 1 and 7.¹⁵

¹² We are unable to apply the same procedure to earnings conference call transcripts because standard data sources do not include conference call transcripts from the pre-shock period.

¹³ Prior studies validate this approach. For instance, [Li et al. \(2021\)](#) validate their corporate culture measure by comparing it to established benchmarks in innovation, integrity, product quality, respect, and teamwork, finding strong positive correlations. Similarly, [An et al. \(2024\)](#) manually code enforcement actions and compare the results with the machine-learning-derived measure, demonstrating consistency between both methods and reinforcing the validity of the machine-learning approach.

¹⁴ We present the 30 most frequently used words and phrases related to supply chain disclosures in [Appendix F](#).

¹⁵ We use the Term Frequency Inverse Document Frequency (TFIDF) index as our weights. This weighting scheme accounts for a word's importance in a given document and in the entire corpus of transcripts. Our inferences are unchanged if we use WFIDF as an alternative weighting scheme. In TFIDF, term frequency is the numerator, whereas, in WFIDF, $\log(1 + \text{term frequency})$ is the numerator.

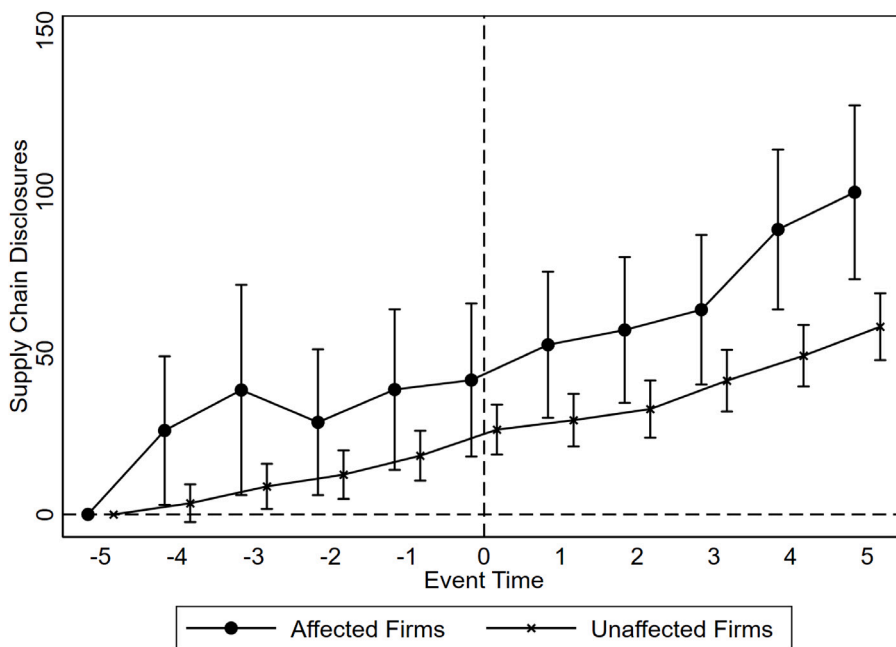


Fig. 6. Trends for supply chain disclosures in event time.

This figure displays the evolution of supply chain disclosures in event time for the sample, separately for affected and unaffected (control) firms. The continuous measure of supply chain disclosures is based on textual analysis of items 1 and 7 of 10-K filings using a dictionary constructed by the *Word2vec* algorithm. The 95% confidence intervals are based on standard errors clustered by firm. Data are for the years 1994–2004.

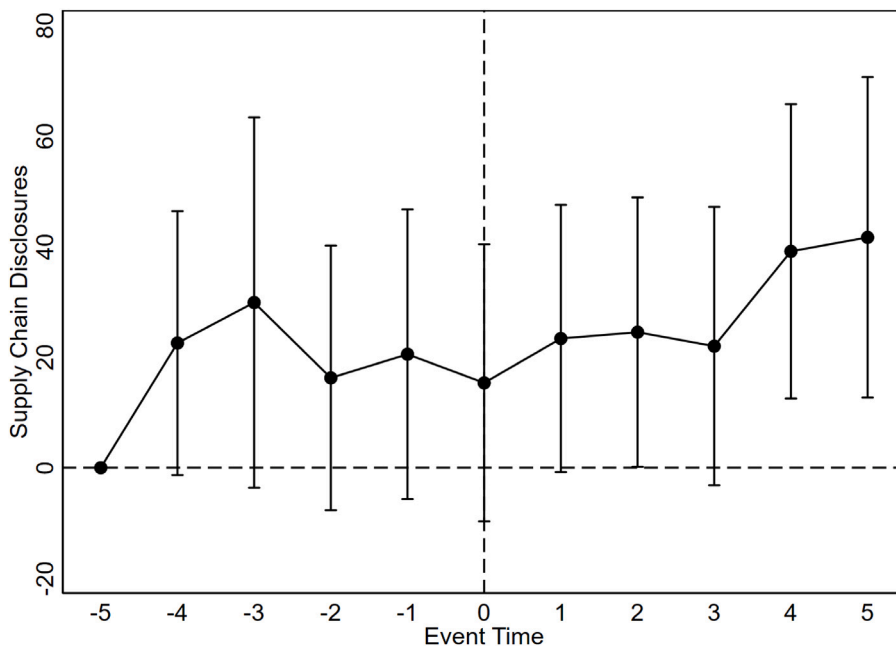


Fig. 7. Difference-in-differences trends for supply chain disclosures in event time.

This figure displays the evolution of supply chain disclosures in event time for the sample. The continuous measure of supply chain disclosures is based on textual analysis of items 1 and 7 of 10-K filings using a dictionary constructed by the *Word2vec* algorithm. The figure shows trends for the affected sample, relative to the control group of unaffected firms. The 95% confidence intervals are based on standard errors clustered by firm. Data are for the years 1994–2004.

Fig. 6 shows the trendlines for our measure of supply chain disclosures separately for affected and control firms, with 1994 as the benchmark period. The plot shows that all firms have consistently increased their disclosures related to their supply chain over time. Fig. 7 shows difference-in-differences trends for affected relative to control firms and shows similar levels of supply chain disclosures for both groups in the post shock period relative to the year of the shock.

Our measure *Low Supply Chain Disclosures* is an indicator for below median values of the supply chain disclosure measure created using *Word2Vec*. A possible concern with this measure of supply chain disclosures is that some managers who do not disclose much might be motivated by trying to avoid proprietary costs of disclosure. If, however, certain non-disclosers are indeed as informed as those that disclose (but choose to withhold information due to proprietary costs), such measurement error would make it more difficult to identify any significant effect of reporting quality on TFP.

We present Pearson and Spearman correlations for the various measures of reporting quality in Table 7, Panel A. Whereas *Class Action Lawsuit*, *SEC Investigation*, and *Restatement* show fairly substantial positive correlations with each other, *Weak Disclosure Controls* and *Low Supply Chain Disclosures* have small negative correlations with the remaining reporting quality measures.

While our measures primarily reflect the quality of external reporting, our focus is on assessing information that ultimately can aid in improving internal decisions. This could include internal (managerial) reporting that is used directly for internal decisions or external (financial) reporting that aids outsiders in monitoring management and pushing managers to enhance firm value. Our measures based on external reporting quality likely also capture variation in internal information flows, as previous research highlights a strong connection between financial and managerial accounting systems. For example, Chen et al. (2018) examine conglomerate firms and find that their external communications quality suffers when the information asymmetry between divisional managers and top managers is more severe. Hemmer and Labro (2008) develop a theoretical model to demonstrate that the properties of managerial and financial accounting systems are not independent. Gallemore and Labro (2015) use measures of external reporting quality as proxies for internal information quality. They find that firms with higher scores on information quality metrics exhibit a greater capacity to avoid taxes, consistent with the notion that the quality of the internal information used for tax planning decisions determines firms' ability to avoid taxes. Other work supporting the link between financial and managerial accounting systems includes Dichev et al. (2013) and Kaplan (1984).

To assess whether firms that score higher on pre-shock accounting and disclosure quality have greater increases in productivity following the shock, we estimate variations of the following OLS model:

$$\begin{aligned} TFP_{it} = & \theta_0 + \theta_1 Affected_i + \theta_2 Post_t + \theta_3 Low\ Reporting\ Quality_i \\ & + \theta_4 Affected_i \times Post_t + \theta_5 Low\ Reporting\ Quality_i \times Post_t \\ & + \theta_6 Affected_i \times Low\ Reporting\ Quality_i \times Post_t + \gamma X_{it-1} + \alpha_i + \delta_t + \epsilon_{it} \end{aligned} \quad (6)$$

where *Low Reporting Quality* represents our measures of accounting and disclosure quality discussed above. The remaining variables are as defined before. The main effects on *Affected*, *Low Reporting Quality*, and *Post* are subsumed by firm and year fixed effects. We expect θ_4 to be positive and significant if high-quality reporting is associated with increases in firm productivity. We expect θ_6 , which captures the incremental effect of low reporting quality on TFP, to be negative.

The results from our estimation are presented in Table 7, Panel B.¹⁶ Across all columns, we find that reporting quality influences the relation between affected firms and TFP in the post-shock period. In column (1) of the table, the measure of reporting quality is whether the firm was involved in a securities class action or collective action lawsuit related to financial reporting prior to the shock. When reporting quality is high (i.e., *Class Action Lawsuit* = 0), affected firms witness a 12.6% increase in post-shock productivity relative to unaffected firms. However, when reporting quality is low (*Class Action Lawsuit* = 1), the incremental change in the TFP of affected firms is significantly lower. Specifically, the interaction term *Affected* $\times$ *Low Reporting Quality* $\times$ *Post* is negative and statistically significant, indicating that the relative post-shock productivity gain for affected firms is 7.8% lower when reporting quality is low. The sum of coefficients test further confirms that the total effect for low-reporting quality firms (0.038) is smaller than that for high-reporting quality firms (0.119) and not statistically different from that of unaffected firms (*F*-statistic = 1.13, *p*-value = 0.2891). We find consistent results in column (2), where our measure of reporting quality is *Weak Disclosure Controls*.

In columns (3) and (4), we find that firms with high reporting quality (i.e., *SEC Investigation* = 0 or *Restatement* = 0) witness an 11.3% increase in TFP. The incremental change in TFP for firms with low reporting quality (*SEC Investigation* = 1 or *Restatement* = 1) is 6% lower. However, low-reporting quality firms see an increase in productivity relative to unaffected firms, but of a lower magnitude. Specifically, these firms see a 4.6% increase in TFP and as the *F*-statistics show, this increase is significantly different from zero.¹⁷

In column (5), reporting quality is based on the supply chain disclosure measure created using the *Word2vec* algorithm. When reporting quality is high (i.e., *Low Supply Chain Disclosures* = 0), relative to unaffected firms, affected firms witness a 12.2% increase in TFP, whereas when reporting quality is low, the incremental change in TFP is 6.8% lower. The sum of coefficients test indicates that low-reporting quality firms experience a significant 5% increase in TFP relative to unaffected firms. Overall, the results in Table 7, Panel B, show that affected firms with better accounting and disclosure practices increase their TFP more in the post-shock period.¹⁸

¹⁶ We report the full Table 7, Panel B, results including coefficients on the control variables, in Table IA1 of Appendix G.

¹⁷ These magnitudes are calculated as follows: $(e^{0.107-0.062} - 1) \times 100 = 4.6$. This is illustrative of the approach we use in all subsequent analogous cases.

¹⁸ These magnitudes cannot be strictly compared to prior work which primarily studies the relation between accounting quality and industry dispersion in productivity, rather than firm-specific changes in productivity. Nonetheless, for comparison, Hann et al. (2020) find that moving from the bottom to the top decile of accounting informativeness is associated with 8%–10% decrease in the 90–10 percentile range of TFP across firms in an industry, and 10%–13% decrease in the within-industry standard deviation. Similarly, Barrios et al. (2019) find that variation in financial reporting quality explains 10%–20% of intra-industry dispersion in TFP.

In a robustness test, we re-estimate the models in Table 7, Panel B, while controlling for a measure of managerial ability (Demerjian et al., 2012), and find similar results (Table 7, Panel C). Demerjian et al. (2012) use data envelopment analysis (DEA) to create their measure of managerial ability. DEA assesses how efficiently managers convert firm resources into revenues relative to their industry peers by creating an efficient frontier, with firms on this frontier scoring highest in efficiency, and others scoring lower based on their distance from the frontier. The DEA scores reflect both managerial ability and firm-specific factors, thus the scores are adjusted by controlling for the firm-specific factors to isolate the “managerial ability” component. After controlling for the Demerjian et al. (2012) measure, the remaining variation in our reporting quality measures is that which is not associated with managerial ability. Our reporting quality measures also show low correlation with the managerial ability measure, suggesting that they capture distinct aspects of reporting quality separate from managerial ability. The correlation coefficients range from -0.0593 for *Low Supply Chain Disclosures* to 0.0077 for *Restatement*, and are generally not statistically significant at conventional levels.

Overall, the results in this section suggest that the quality of accounting information and disclosure practices is related to changes in firms’ productivity following the supply chain shock. That is, higher reporting quality aids firms in taking advantage of the opportunity to cut slack and improve their productivity.

6. Channels linking reporting quality to productivity

In our next set of analyses, we relate the changes in the outcomes detailed in Section 5.1 to reporting quality and TFP. In particular, we examine whether the relation that our measures of accounting and disclosure quality exhibit with TFP is more pronounced among firms that incur post-shock restructuring charges or make bigger cuts in capital expenditures, R&D and SG&A. Firms most affected by the shock would be most incentivized to fully utilize their available information to enhance productivity levels.

We present these analyses in Table 8, where we split the sample into firms that restructured or cut investments following the shock (Table 8, Panel A) and those that did not (Table 8, Panel B). Most firms in our sample took such actions, including 89% of affected firms and 74% of unaffected firms. The results in Table 8 confirm that TFP increases for affected firms with higher accounting and disclosure quality are concentrated among firms that restructure or cut investments following the shock.

Panel A (sample of firms that restructure or cut investments) shows that when reporting quality is high, affected firms experience an 11.7–14.1% increase in TFP relative to unaffected firms in the period following the shock. However, when reporting quality is low, affected firms witness incrementally lower TFP, which is significantly lower in all cases except in column (4), where the measure of low reporting quality is *Restatement*. Also, the sum of coefficients test shows that the total effect for low-reporting quality firms is not significantly different from zero.

Panel B presents results for the sample of firms that do not restructure or cut investments following the shock. The table shows that these firms, even if they have higher accounting quality and better disclosures, see no increases in TFP following the shock.¹⁹ However, we are unable to estimate the model using the reporting quality measures of *Weak Disclosure Controls*, *SEC Investigation*, and *Restatement* due to the small sample size. Overall, the findings presented in Table 8 indicate that superior accounting quality and enhanced disclosures support firms in boosting productivity when they undertake additional measures such as restructuring or scaling back investments.

Our results are consistent with higher reporting quality facilitating increases in productivity following a supply chain shock. Our measures of reporting quality aim to capture how good the information collection and dissemination systems of a firm are. In turn, the information obtained and communicated from such systems helps managers direct resources to their most efficient and effective utilization during the process of restructuring to respond to the shock. Our results thus highlight the critical function of accounting information and its disclosure in aiding managerial decision-making, especially under stress, and suggest that accounting information can be a crucial determinant of firm resilience and adaptability in the face of unexpected external shocks.

7. Reporting quality and post-shock performance

Finally, we examine whether firms with higher-quality accounting and disclosure practices are better equipped to navigate competitive pressures after experiencing the supply chain shock, given that such firms are better able to implement actions that increase TFP. To assess whether such firms perform better in the aftermath of the crisis, we estimate variations of the following OLS model:

$$\begin{aligned} Performance_{it} = & \theta_0 + \theta_1 Affected_i + \theta_2 Post_t + \theta_3 Low\ Reporting\ Quality_i \\ & + \theta_4 Affected_i \times Post_t + \theta_5 Low\ Reporting\ Quality_i \times Post_t \\ & + \theta_6 Affected_i \times Low\ Reporting\ Quality_i \times Post_t + \alpha_i + \delta_t + \epsilon_{it} \end{aligned} \quad (7)$$

¹⁹ We confirm that a lack of power is not an issue in these tests. We conducted a power analysis using the adjusted R-squared values from our regression models as an indicator of effect size (Cohen, 2013). Specifically, we compared a full model that includes the interaction terms related to the intervention with a reduced model that omits these terms, maintaining identical control variables and fixed effects in both. By using the adjusted R-squared values from these models, we estimated the sample size required to achieve adequate power (0.80) to detect a meaningful difference between the models. We find that the minimum number of observations that would be required in columns (1), (2), and (3) of Panel B are 508, 259, and 1089. Our actual number of observations is 1754 in columns (1) and (3), and 990 in column (2).

Table 8
Reporting quality and total factor productivity (in subsamples).

Panel A: Sample of firms that restructure or cut investments following the shock					
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Weak Disclosure Controls	Low Reporting Quality = SEC Investigation	Low Reporting Quality = Restatement	Low Reporting Quality = Low Supply Chain Disclosures
	TFP	TFP	TFP	TFP	TFP
	(1)	(2)	(3)	(4)	(5)
Affected $\times$ Post	0.123*** (4.777)	0.113*** (4.367)	0.111*** (4.503)	0.112*** (4.564)	0.132*** (4.284)
Low Reporting Quality $\times$ Post	0.027 (0.837)	-0.097*** (-3.638)	-0.034** (-1.978)	0.038 (0.379)	0.003 (0.147)
Affected $\times$ Low Reporting Quality $\times$ Post	-0.135** (-2.371)	-0.080** (-1.979)	-0.076** (-2.388)	-0.147 (-1.534)	-0.118*** (-3.075)
Sum of coefficients (<i>F</i> -statistic)	-0.012 (0.05)	0.033 (1.13)	0.035 (2.71)	-0.035 (0.14)	0.014 (0.32)
Observations	5398	4261	5398	5398	3284
Adjusted R ²	0.859	0.855	0.858	0.858	0.798
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

Panel B: Sample of firms that do not restructure or cut investments following the shock		
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Low Supply Chain Disclosures
	TFP	TFP
	(1)	(2)
Affected $\times$ Post	-0.062 (-1.587)	0.010 (0.343)
Low Reporting Quality $\times$ Post	-0.065* (-1.806)	0.011 (0.435)
Affected $\times$ Low Reporting Quality $\times$ Post	0.265* (1.785)	0.191 (0.988)
Observations	1754	990
Adjusted R ²	0.794	0.788
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes

This table presents OLS estimates of changes in TFP for firms adversely affected by the earthquake (*Affected*) relative to those that are not. [Panel A](#) includes the sample of firms that restructure or cut investments following the shock, whereas [Panel B](#) includes the sample of firms that do not take such actions. The indicator variable *Post* takes the value of 1 for 1999–2004. The dependent variable (TFP) is calculated based on [Levinsohn and Petrin \(2003\)](#). Controls are lagged by one year and include: *Acquisitions*, *Capital expenditures*, *Cash holdings*, *Inventory*, *Leverage*, *Margin(gross)*, *Net debt*, *Net equity*, *R&D*, *ROA*, *Sales*, *SG&A* and *Size*. The *F*-statistic tests if the sum of coefficients on *Affected* $\times$ *Post* and *Affected* $\times$ *Low Reporting Quality* $\times$ *Post* equals zero. Standard errors are clustered at the firm level. To mitigate the effects of extreme observations, all continuous variables are winsorized at the 1% and 99% tails of their respective distributions in each sample year. All variables are defined in [Appendix C](#). The *t*-statistics are presented in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

where *Performance* represents firm performance and is one of three outcomes: return on assets (*ROA*), gross margins (*Margin(gross)*), and the market share of sales (*Market Share*). *ROA* for each firm *i* in year *t* is *i*'s net income in year *t* scaled by the average of firm *i*'s year *t* and year *t* – 1 total assets. *Margin(gross)* is firm *i*'s revenue minus cost of goods sold in year *t* scaled by *i*'s revenue in year *t*. Finally, *Market Share* is the percentile rank of firm *i*'s sales in year *t* scaled by total sales of all firms in the industry in year *t*. We take percentile ranks of market shares as this variable is highly skewed. The remaining variables are as defined before. The main effects on *Affected*, *Low Reporting Quality*, and *Post* are subsumed by the inclusion of firm and year-fixed effects. We expect θ_4 in Eq. (7) to be positive and significant if firms with higher-quality accounting and disclosure practices witness post-shock increases in firm performance. The incremental effect on post-shock performance for firms with low reporting quality is captured by θ_6 .

We present results from the estimation of Eq. (7) in [Table 9](#) for our various firm performance and reporting quality measures. In [Panel A](#) of the table, the measure of firm performance is *ROA*. The panel shows that when reporting quality is high (*Low Reporting*

Quality = 0), affected firms witness an 8.4%–11.2% increase in post-shock return on assets relative to unaffected firms. When reporting quality is low (*Low Reporting Quality* = 1), we find that in columns (1), (2), and (5), the change in *ROA* of affected firms is not significantly different from that of unaffected firms in the post-shock period (as inferred from test of whether the sum of coefficients on *Affected* × *Post* and *Affected* × *Low Reporting Quality* × *Post* equals zero).

In columns (3) and (4), firms with low reporting quality see a decrease in performance relative to unaffected firms. Specifically, firms with low reporting quality see a 16.9%–40.8% decline in return on assets, and as the *F*-statistics show, this decline is significantly different from zero. These results highlight the important role of firms' reporting quality in managing major operational disruptions and shaping future firm performance.

Our results in [Panel B](#) and [Panel C](#), which use gross margins and market share as measures of performance, are broadly consistent with the results in [Panel A](#). Affected firms with high reporting quality witness a 14.9%–20% (4.9%–6.7%) increase in gross margins (market share) post-shock relative to unaffected firms. These findings confirm that affected firms with higher reporting quality exhibit stronger performance after the crisis.

We also plot the dynamic post-shock performance for affected firms with high-quality reporting systems (i.e., the coefficients of *Affected* × *Post* from [Table 9](#), [Panels A–C](#)).²⁰ The event study trends for *ROA*, *Margin (gross)* and *Market Share* indicate that firms with higher financial reporting quality experienced significant and sustained performance improvements following the supply chain

²⁰ These plots are presented in Figure IA1–Figure IA3 of Appendix G.

Table 9
Reporting quality and firm performance measures.

Panel A: ROA as the measure of firm performance					
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Weak Disclosure Controls	Low Reporting Quality = SEC Investigation	Low Reporting Quality = Restatement	Low Reporting Quality = Low Supply Chain Disclosures
	ROA	ROA	ROA	ROA	ROA
	(1)	(2)	(3)	(4)	(5)
Affected × Post	0.112*** (3.584)	0.103*** (3.361)	0.109*** (3.810)	0.105*** (3.659)	0.084** (2.342)
Low Reporting Quality × Post	0.023 (0.567)	−0.195 (−0.762)	0.271* (1.864)	0.027 (0.299)	−0.031 (−0.973)
Affected × Low Reporting Quality × Post	−0.100* (−1.689)	0.265 (1.033)	−0.517*** (−3.504)	−0.274*** (−2.868)	−0.014 (−0.257)
Sum of coefficients (<i>F</i> -statistic)	0.012 (0.06)	0.368 (2.08)	−0.408*** (7.94)	−0.169* (3.44)	0.070 (2.65)
Observations	10,938	8359	10,938	10,938	5372
Adjusted R ²	0.608	0.616	0.608	0.608	0.516
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

Panel B: Gross margins as the measure of firm performance					
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Weak Disclosure Controls	Low Reporting Quality = SEC Investigation	Low Reporting Quality = Restatement	Low Reporting Quality = Low Supply Chain Disclosures
	Margin (gross)	Margin (gross)	Margin (gross)	Margin (gross)	Margin (gross)
	(1)	(2)	(3)	(4)	(5)
Affected × Post	0.150** (1.966)	0.200*** (3.515)	0.149** (2.146)	0.149** (2.140)	0.139 (1.016)
Low Reporting Quality × Post	0.133*** (2.911)	0.072 (0.373)	0.117** (2.453)	0.091* (1.764)	0.064 (0.578)
Affected × Low Reporting Quality × Post	−0.095 (−1.077)	−0.058 (−0.295)	−0.345** (−2.415)	−0.319** (−2.196)	−0.019 (−0.131)
Sum of coefficients (<i>F</i> -statistic)	0.055 (1.45)	0.142 (0.57)	−0.196 (2.54)	−0.170 (1.86)	0.120* (3.85)
Observations	10,725	8180	10,725	10,725	5303
Adjusted R ²	0.359	0.358	0.359	0.359	0.227
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

(continued on next page)

Table 9 (continued).

Panel C: Market share as the measure of firm performance					
	Low Reporting Quality = Class Action Lawsuit	Low Reporting Quality = Weak Disclosure Controls	Low Reporting Quality = SEC Investigation	Low Reporting Quality = Restatement	Low Reporting Quality = Low Supply Chain Disclosures
	Market Share	Market Share	Market Share	Market Share	Market Share
	(1)	(2)	(3)	(4)	(5)
Affected $\times$ Post	6.734*** (3.629)	6.269*** (3.358)	6.208*** (3.538)	6.181*** (3.522)	4.883** (2.032)
Low Reporting Quality $\times$ Post	-0.135 (-0.089)	4.989 (0.854)	0.587 (0.170)	-0.937 (-0.312)	-1.412 (-1.054)
Affected $\times$ Low Reporting Quality $\times$ Post	-6.254 (-1.233)	3.605 (0.589)	-16.262*** (-3.319)	-14.740*** (-3.208)	-4.575 (-1.251)
Sum of coefficients (<i>F</i> -statistic)	0.480 (0.01)	9.874* (2.87)	-10.054** (4.83)	-8.559** (4.06)	0.308 (0.01)
Observations	10,962	8365	10,962	10,962	5377
Adjusted R ²	0.905	0.904	0.905	0.905	0.901
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

This table presents OLS estimates of changes in firm performance measures for firms adversely affected by the earthquake (*Affected*) relative to those that are not. In Panel A, our measure of performance is return on assets (*ROA*), whereas in Panel B and Panel C, it is gross margins (*Margin(gross)*) and percentile rank of market share of sales (*Market Share*), respectively. The indicator variable *Post* takes the value of 1 for 1999–2004. The *F*-statistic tests if the sum of coefficients on *Affected* $\times$ *Post* and *Affected* $\times$ *Low Reporting Quality* $\times$ *Post* equals zero. Standard errors are clustered at the firm level. To mitigate the effects of extreme observations, all continuous variables are winsorized at the 1% and 99% tails of their respective distributions in each sample year. All variables are defined in Appendix C. The *t*-statistics are presented in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-tailed).

shock. Before the shock ($t_0 - 5$ to $t_0 - 1$, where t_0 is the *Event Time*), all three metrics remain largely stable, suggesting no significant pre-trend differences. However, beginning at t_0 , ROA increases sharply, peaking around $t_0 + 1$. This suggests that firms with better reporting quality quickly improved asset utilization efficiency, likely by boosting TFP. Gross margins also rise following the shock, though the increase is more uneven and less immediate, reflecting the time required for any adjustments to take effect. Market share shows a sharp initial increase at $t_0 + 1$ that remains sustained over the following years, suggesting that firms with superior financial reporting leveraged their information advantage to make better strategic decisions and gain a competitive edge. These trends support the argument that firms with high-quality reporting were able to respond more effectively to the shock and were able to improve their profitability and competitive position over time, even considering that managerial actions may require some time to pass before producing effects.

8. Conclusion and discussion

We investigate the role of reporting quality in how productively firms restructure in the face of a supply chain disruption. Prior work shows that reporting quality matters for day-to-day operating decisions and less frequent, but recurring, decisions such as major investments. When stable regimes are disrupted, however, high-quality information may be difficult to come by and be especially valuable. Using several measures of reporting quality, we find that higher reporting quality is linked to bigger increases in productivity following the shock.

We use the 1999 Taiwan earthquake as a supply chain shock that is unrelated to pre-shock reporting quality but incentivizes firms to increase productivity. The quake adversely affected US high-tech firms that sourced components from Taiwan. As a result, input prices increased for the affected firms but not for their competitors sourcing from outside Taiwan. Affected firms could not pass these costs onto consumers due to the highly competitive nature of the industry and, consequently, might have restructured their operations in response. Thus, we first examine changes in the operating and investing activities of affected firms compared to a control group of unaffected firms. We find that affected firms see a post-shock increase in restructuring costs, and reductions in capital expenditures, R&D, and SG&A, with no significant changes in acquisitions or employee numbers.

Disruptive shocks can reduce managerial slack and organizational inertia, and incentivize firms to increase productivity. Thus, we next investigate whether the shock led to an increase in productivity at affected firms. Using TFP to measure productivity and a difference-in-differences design, our results show that TFP increases for affected firms relative to control firms following the earthquake. We also find that such increases are concentrated among firms that incurred restructuring charges and cut investments. From these baseline findings, we test and find that the post-quake increase in TFP is significantly greater for affected firms with higher reporting quality.

We further investigate the mechanism by relating changes in restructuring and investing to reporting quality and TFP. We find post-shock improvements in TFP are strongly linked to higher reporting quality; however, these advantages manifest only in firms

that engage in restructuring or investment reduction after the shock. These results imply that high-quality reporting, which likely supports better strategic decision-making and efficient resource management, is vital for firms that adjust their operations after the shock. Finally, we find that affected firms with higher reporting quality exhibit stronger performance after the crisis.

Our findings suggest significant frictions impede managers from voluntarily improving their accounting systems to the extent needed to obtain the maximum productivity benefits, at least during times when the firm is reacting to a major change in circumstances. One such friction could be that the cost of improving reporting quality may exceed the benefits. Alternatively, managers could lack awareness of, or have high uncertainty about, the full benefits of improving their reporting quality. Finally, accounting systems may exhibit inertia due to individual behavioral biases favoring the status quo and an emphasis on consistency.

If lack of awareness of, or uncertainty about, the benefits is an important friction restraining firms from voluntarily improving the quality of their accounting information, our evidence on the productivity impact demonstrates a key benefit. If costs exceeding benefits in some circumstances is another important friction, our findings help to identify restructurings in response to negative shocks as situations where the benefits are likely to exceed the costs. For high-tech firms faced with a supply chain shock, the value of reporting quality is critical for firms that reduce investment or restructure. In other settings, reporting information related to growth options rather than abandonment options may be more important.

Future research could investigate the extent to which high costs, lack of awareness of benefits, uncertainty about benefits, individual behavioral biases favoring the status quo, or overemphasis on consistency in accounting and auditing practices impede voluntary improvements in firms' reporting quality. The findings from such work could improve our understanding of why reporting quality varies considerably even among firms in the same industry. Such knowledge is relevant both for better informing firms' voluntary reporting quality choices and as an input into debates about mandated changes in accounting.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Semiconductor imports by the US, by source location

	Number of units (In millions)		Share of total		Share of change 1998–99
	1998	1999	1998	1999	
Malaysia	3620	4788	22.1%	24.7%	38%
Japan	2366	2945	14.5%	15.2%	19%
Taiwan	2193	2343	13.4%	12.1%	5%
Philippines	2175	2616	13.3%	13.5%	15%
South Korea	1476	1690	9.0%	8.7%	7%
Rest of the world	4524	5016	27.7%	25.9%	16%
Total semiconductor imports	16,353	19,398			

This table shows semiconductor imports, in number of units, by the US in the years 1998 and 1999. The data are sourced from the UN Comtrade database and are for the commodity code 8542 (Electronic integrated circuits and microassemblies).

Appendix B. Sample description

We search SEC filings as well as the Factset database for information related to suppliers of US high-technology manufacturing firms prior to the earthquake. Factset consolidates firm-level information on supply-chain relationships from regulatory filings and company websites. We search SEC filings for variations of the following words: *supply*, *supplier*, *procurement*, *seller*, *foundry*, *Taiwan* and *Taiwanese*. We read the filings to ascertain whether the supplier is located in Taiwan. Often, firms name their largest suppliers without specifying a location. We compile names of Taiwanese suppliers from industry reports and plot supplier locations on a map showing earthquake intensity (Fig. 1). Taiwan Semiconductor Manufacturing Corporation (TSMC) and United Microelectronics Corporation (UMC) are the largest foundries in Taiwan, and several US firms source from these foundries. A description of the three-digit SIC codes for firms included in the sample are given below. These industries are a subset of industry number six (computers, software, and electronic equipment) in the Fama–French 12 industry classification.

Industry description for firms with suppliers in Taiwan.			
SIC code	Description	Number firms	Percent firms
357	Computer and office equipment	19	18%
366	Communications equipment	13	13%
367	Electronic components and accessories	70	68%
382	Laboratory apparatus and analytical, Optical, Measuring, and Controlling instruments	1	1%
		103	100%

Appendix C. Variable definitions

Variable	Definition	Source
Acquisitions	Acquisition related cash outflows (funds used for and costs related to acquisitions) scaled by average assets	Compustat, Authors' calculations
Affected	Indicator that takes a value of 1 for firms that source raw materials from Taiwan, and 0 otherwise	Authors' calculations
Capital expenditures	Cash flow related to capital expenditures scaled by average assets	Compustat, Authors' calculations
Cash holdings	Cash and short-term marketable securities scaled by total assets	Compustat, Authors' calculations
Class Action Lawsuit	Indicator that takes a value of 1 if the beginning of the class period in a securities class action or collective action lawsuit related to financial reporting occurred in the five years prior to the shock	Audit Analytics–Legal case
Employees	Number of employees scaled by average assets	Compustat, Authors' calculations
Inventory	Total inventory (raw materials, work-in-progress, and finished goods) scaled by total assets	Compustat, Authors' calculations
Leverage	Debt in current liabilities plus total long-term debt scaled by total assets	Compustat, Authors' calculations
Low Supply Chain Disclosures	Indicator for below median values of <i>Supply Chain Disclosures</i> in the year before the shock	10-K filings, authors' calculations
Margin (gross)	Revenue minus cost of goods sold scaled by revenue	Compustat, Authors' calculations
Market share	Percentile rank of market share of sales	Compustat, Authors' calculations
Net debt	Debt issuance less debt retirement scaled by average assets	Compustat, Authors' calculations
Net equity	Equity sales less equity purchases scaled by average assets	Compustat, Authors' calculations
Post	Indicator variable that takes a value of 1 for the years 1999–2004, and 0 otherwise	Authors' calculations
Restructure	Indicator variable equal to 1 if at least one of Compustat annual data items RCP, RCA, RCEPS, RCD is positive, and 0 otherwise	Compustat, Authors' calculations
R&D	Research and development expense scaled by average assets	Compustat, Authors' calculations
Restatement	Indicator which equals 1 if the firm had an intentional misstatement in the years prior to the shock	Hennes et al. (2008)
ROA	Return on assets: Net income scaled by average assets	Compustat, Authors' calculations
Sales	Total sales scaled by average assets	Compustat, Authors' calculations
SEC Investigation	Indicator that takes a value of 1 if the SEC started an investigation related to financial reporting or disclosure issues against the firm in the 5 years prior to the shock.	FOIA requests (Blackburne et al., 2021 ; Holzman et al., 2024)
SG&A	Selling, general, and administrative expense scaled by average assets	Compustat, Authors' calculations

Variable	Definition	Source
Size	Natural log of total assets	Compustat, Authors' calculations
Supply Chain Disclosures	Extent of supply chain related disclosures in firms' 10-K filings created using <i>Word2vec</i>	10-K filings, authors' calculations
TFP	The natural logarithm of one plus total factor productivity estimated based on Levinsohn and Petrin (2003)	Compustat, NBER-CES Manufacturing Industry Database, Authors' calculations
Unaffected	Indicator that takes a value of 1 for firms that did not source raw materials from Taiwan, and 0 otherwise	Authors' calculations
Weak Disclosure Controls	Indicator that takes a value of 1 if the firm was revealed to have weak disclosure controls under Section 302 of SOX in the 5 years prior to the shock	Audit Analytics

Appendix D. TFP estimation

To estimate TFP, we assume a Cobb–Douglas production function, which in log-linear form is as follows:

$$y_t = \beta_0 + \beta_l l_t + \beta_k k_t + \beta_m m_t + \omega_t + \eta_t, \quad (\text{D.1})$$

where y_t is the natural logarithm of a firm's output measured as gross revenue; l_t is the natural logarithm of the input labor; k_t is the natural logarithm of capital; and m_t is the natural logarithm of the intermediate input (materials). The error consists of ω_t , which represents unobserved productivity shocks that impact the firm's choice of inputs, and η_t , which is the error that is uncorrelated with input choices. Unobserved productivity shocks can simultaneously affect both the error term and the inputs, thus OLS estimates will be biased. The GMM estimator developed by [Levinsohn and Petrin \(2003\)](#) uses intermediate goods as proxies for the unobservable productivity shocks, based on the argument that intermediate goods respond more smoothly to productivity shocks.²¹

We measure the output y as the logarithm of total sales deflated by a price index; labor l as the logarithm of the total number of employees times the average wage in an industry-year; and m as the logarithm of intermediate inputs deflated by a price index. Intermediate inputs are calculated as the sum of cost of goods sold and selling, general, and administrative expenses, minus depreciation expenses. Capital k is estimated as the logarithm of the sum of property, plant, and equipment net of depreciation and the stock of R&D capital, deflated by a price index. Data on price deflators and industry wages are sourced from the NBER-CES Manufacturing Industry Database, which provides these data by year and four-digit SIC industry code.

Firms in the high-technology manufacturing industries have large R&D expenses, so we include the stock of R&D capital in k . We calculate the stock of R&D by using the perpetual inventory method and assuming an amortization rate of 15% ([Lev and Radhakrishnan, 2009](#); [Sliker, 2007](#)). Specifically, the stock of R&D capital (RDA) for firm i at time t is calculated recursively as follows:

$$RDA_{it} = (1 - d) \times RDA_{it-1} + R\&D_{it}, \quad (\text{D.2})$$

where d is the amortization rate and $R\&D$ represents research and development expenses. To implement Eq. (D.2), the initial value of RDA is chosen as follows:

$$RDA_{it=0} = \frac{R\&D_{it=1}}{g + d}, \quad (\text{D.3})$$

where g is the average real growth rate of firm-level R&D expenditures.

In estimating Eq. (D.1), we follow prior work to drop extreme observations from our sample ([Hann et al., 2020](#); [Keller and Yeaple, 2009](#)).²² After removing outliers, our estimated factor elasticities for labor (β_l) and materials (β_m) are 0.280 and 0.408, respectively, and are comparable to those reported in prior work. For example, [Hann et al. \(2020\)](#) report coefficients of 0.249 and 0.380 for labor and materials. [Levinsohn and Petrin \(2003\)](#) report coefficients for labor in the range of 0.051–0.193 and for materials in the range of 0.420–0.550. Our estimate for capital elasticity (β_k) is higher at 0.887 because we include the stock of R&D capital in our estimates, and R&D outlays can significantly impact output in high-technology manufacturing. In comparison, [Hann et al. \(2020\)](#), whose sample consists of all manufacturing industries, report a capital elasticity of 0.275. [Levinsohn and Petrin \(2003\)](#), whose sample consists of food products, metals, textiles, and wood products, report capital elasticities in the 0.180–0.290 range.

²¹ For details related to the estimation, please see [Levinsohn and Petrin \(2003\)](#) and [Petrin et al. \(2004\)](#).

²² We exclude firm-year observations exhibiting extreme fluctuations in key metrics, specifically those with year-over-year changes exceeding 300% increases or 50% decreases in capital, revenue, or materials. Additionally, any firm with a revenue-to-material ratio below one or above ten is omitted, as are cases with negative values for the number of employees, sales, depreciation, capital expenditures, or cost of goods sold. Observations lacking sufficient data for TFP estimation are also dropped. Furthermore, we limit our focus to firms with a minimum size, excluding those with fewer than 30 reported employees.

Appendix E. Empirical framework

Let the productivity of a firm i at time t be represented by Y_{it} . Let A_i represent the quality of firm i 's accounting system, which is predetermined and does not vary with t . There are two time periods, $t = 0$ and $t = 1$. The firm experiences a random productivity shock S at $t = 1$.

First, we assume that even though the quality of the firm's accounting system contributes to firm productivity, it does not affect i 's response to S . In this case, productivity can be written as follows:

$$Y_{it} = \alpha + \beta A_i + \delta X_i + \gamma S_t + \epsilon_{it} , \quad (\text{E.1})$$

where X is a vector of all other firm-specific factors that affect productivity and ϵ is the error term that captures unobserved factors affecting Y . S is exogenous in the sense that it is uncorrelated with ϵ . Thus, γ identifies the effect of the shock on productivity, Y .

Now suppose that the firm's response to the shock S will depend on the predetermined quality of the accounting system A . Thus, the interaction of A with S will determine i 's response to the shock. In this case, productivity is written as follows:

$$Y_{it} = \alpha + \beta A_i + \delta X_i + \gamma S_t + \theta(A_i \times S_t) + \epsilon_{it} . \quad (\text{E.2})$$

Before the shock ($t = 0$), Eq. (E.2) reduces to the following as $S_0 = 0$:

$$Y_{i0} = \alpha + \beta A_i + \delta X_i + \epsilon_{i0} . \quad (\text{E.3})$$

After the shock ($t = 1$), Eq. (E.2) can be written as follows as $S_1 = 1$:

$$Y_{i1} = \alpha + \beta A_i + \delta X_i + \gamma + \theta A_i + \epsilon_{i1} . \quad (\text{E.4})$$

Subtracting Eq. (E.3) from Eq. (E.4):

$$\begin{aligned} \Delta Y_i &= Y_{i1} - Y_{i0} = (\alpha + \beta A_i + \delta X_i + \gamma + \theta A_i + \epsilon_{i1}) - (\alpha + \beta A_i + \delta X_i + \epsilon_{i0}) \\ \implies \Delta Y_i &= \gamma + \theta A_i + (\epsilon_{i1} - \epsilon_{i0}) = \gamma + \theta A_i + v_i , \end{aligned} \quad (\text{E.5})$$

where $v_i = \epsilon_{i1} - \epsilon_{i0}$.

θ captures the heterogeneity in the firm's response to the shock depending on the quality of its accounting system A . The exogeneity of S and the predetermined nature of A allows θ to be identified without bias. The exogeneity of S implies that it is uncorrelated with the error term and that θ can be interpreted as the effect of the firm's ability (based on A) to respond to the shock S . Also, because A is predetermined, it is uncorrelated with changes in unobserved factors between the two periods, i.e., with the error v_i .

Appendix F. Supply chain disclosures dictionary

This section presents the most commonly occurring words and phrases in our supply chain disclosures dictionary.

supply	third_party_manufacturer	outsource
supplier	assembly	subassembly
procure	component	adequate_supply
procurement	customer	component_subassembly
vendor	sole_source_supplier	system_component
contract_manufacturer	component_supplier	assembly_test
supply_chain	source_supply	sub-assembly
subcontractor	sourcing	assemble_test
outside_supplier	assembly_testing	timely_delivery
third_party_supplier	raw_material	sole_supplier

Appendix G. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jacceco.2025.101833>.

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